Adaptive Design of Clustered Experiments

Carlos Gonzalez-Perez*

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Abstract

I study optimal experimental and survey design under clustered sampling and budget constraints. A researcher must decide how many clusters and how many units per cluster to sample in order to minimize the variance of a population average estimator. Although sampling additional clusters reduces statistical dependence between observations, clusters are often costly to recruit or administer. The optimal design therefore depends critically on the intra-cluster correlation, which is typically unknown. I propose a two-stage explore-then-commit algorithm that uses a pilot to learn the intra-cluster correlation and then allocates the remaining budget between units and clusters using plug-in optimal rules. The estimator resulting from this two-stage design achieves asymptotically negligible excess variance relative to an oracle design that knows the correlation ex-ante. I further show that these guarantees are minmax optimal up to logarithmic factors and strictly improve upon any static policy that inputs a misspecified correlation. Then I study an adaptive version of the algorithm where the length of the pilot is determined through a data-dependent stopping rule. Finally, under a symmetry condition on the sampling errors, I show that these adaptive designs do not invalidate inference: the resulting estimator is asymptotically independent of the design, therefore the analyst can use all the data from the pilot, and the estimator remains asymptotically normal. These results provide a foundation for adaptive cluster allocation in multistage surveys and stratified randomized controlled trials.

1 Introduction

Over the last few years, surveys and Randomized Control Trials (RCT) have become increasingly larger, not only in size but in scope, spanning several physical and figurative regions (Muralidharan and Niehaus (2017)).[1] This trend is prevalent across the board in survey design and development economics including education, public health, finance, trade, gender and more (Duflo and Banerjee (2017)). There are several reasons behind this trend, most notably, the quest for external validity and efficiency (Muralidharan and Niehaus (2017)). However, when running surveys and experiments on a budget, high fixed cluster costs may limit the scope of the design.

Consider the survey in Sedgwick (2015), in which in-person interviews were conducted to learn

*University of Oxford, Department of Economics. Email: carlos.gonzalezperez@economics.ox.ac.uk. I would like to thank Maximilian Kasy, Davide Viviano, Rahul Singh and participants at the Econometrics Seminar at Harvard for feedback on this manuscript.

[1]Examples of physical regions include sampling from different countries, provinces, districts, or villages, while examples of figurative regions include sampling from different demographic strata, markets, industries, etc.

about the preferences of the British household about the use of medical data. To minimize travel costs, the designer randomly selected a few postcodes (proportionally their size) and then interviewed a few households from each postcode. We encounter similar examples in the experimental design and Development Economics literature. In Mbiti et al. (2019), the authors deploy an experiment at the school level consisting of a combination of school grants and teacher incentives. The implementation featured random selection of 10 districts from mainland Tanzania and random treatment allocation to 35 schools in each district. Both of these designs are examples of **multi-stage** or stratified surveys/experiments. While the resulting estimators remain unbiased, due to random sampling at every stage, their variance depend critically on the design and the distributional assumptions on the population.[2]

The inclusion of multiple clusters is often desirable from a statistical perspective. When observations (or treatment effects) are highly correlated within clusters, the researcher has an incentive to sample individuals from several clusters to minimize the variance of the resulting estimators. On the other hand, the inclusion of additional clusters is often costly, especially in developing countries. Besides travel costs, multi-cluster implementation usually involves reaching agreements with local authorities, training of additional staff, increased coordination efforts, or location-specific knowledge.[3] Importantly, these costs are usually *sunk* and take place before the actual implementation of the experiment/survey. As a result, the optimal *depth* (number of units per cluster) vs *width* (number of clusters) design must trade-off between statistical accuracy and costs. To solve for the optimal design, the analyst must properly account for the cost structure and the degree of **intra-cluster correlation (ICC)**. Intuitively, the marginal value of an additional observation (in terms of variance reduction) from the same cluster is lower whenever the ICC is higher. As a result, in the presence of high intra-cluster correlation, the designer faces an additional incentive to sample from several clusters as opposed to sample many units from the same cluster, i.e. she will favor a *wide* design. Conversely, in the presence of low ICC, the designer may favor *deep* designs, i.e. few cluster with many units per cluster.[4]

While the cost structure may be known to the designer, it is often difficult to make strong assumptions on the share of ICC, especially given its large variability across regions and domains. Moreover, the optimal design is very sensitive to this parameter (see Figure 1), what makes the imputation of the ICC a first order concern in the depth-vs-width problem. For instance, Knox and Chondros (2004) speaks directly about this problem in medical studies. In their work, the authors show that ICC estimates on demographic covariates range from 0.055 to 0.451 and ICC estimates on morbidity covariates range from 0.005 to 0.059 (an order of magnitude variation).

The estimation of the ICC is not only empirically relevant but also challenging. To see why, observe that the ICC is (essentially) a ratio of the within-cluster variance to the between-cluster variance, i.e. how much individuals vary across clusters compared to much they vary across the entire population. Thus, to learn the ICC one needs to estimate the between-cluster variance by sampling

[2]We see this stratified design again and again both in surveys (surveys on CEOs across different industries Datta and Rajagopalan (1998), or land surveys across different regions Lin and Ho (2003)) and RCTs (treatment allocated to public clinics across different districts Karing (2024), to villages across different wards Heß et al. (2021), or to districts across different provinces Ashraf et al. (2020)).

[3]For instance, Meriggi et al. (2024) need to train different social-mobilization teams in rural Sierra Leone, and Li et al. (2026) need to build an algorithm around the hiring infrastructure of a big company. These costs are nearly invariable to the number of treated units in each region/company, and must be covered before the actual implementation of the experiment.

[4]In the extreme case where observations are perfectly correlated within clusters, the analyst extracts all the cluster information by sampling a single individual. In this extreme setting, the analyst would favor very wide designs, i.e. sampling a single unit from as many clusters as possible.

from many clusters—which is, precisely, the expensive element of the design. This dynamic places the designer at a conundrum: Optimal design identification requires knowing the ICC but ICC estimation requires sampling many clusters. Because cluster exploration often requires bearing fixed costs that cannot be undone, ICC estimation damages optimal design implementation.

To the author's knowledge, this is the first paper to investigate the depth-vs-width problem with unknown intra-cluster correlation. To do so, I consider a minimal model with nested errors, linear costs and a hard budget. I then characterize the oracle policy (the variance minimizing design) by solving the designer's problem under known ICC. Secondly, I develop a two-stage algorithm with an explore-then-commit structure, which solves the problem efficiently without any previous knowledge on the intra-cluster correlation. In the exploration stage, the algorithm runs a pilot to learn the ICC under a worst-case approach by sampling as few units and as few clusters as one could hope. In the exploitation stage, the designer allocates the remaining budget between units and clusters according to the optimal design prescribed by the estimated ICC coefficient. I refer to this algorithm as Explore-then-Commit Cluster Selection (`EtC-CS`).

In terms of performance, I first show that the variance of the oracle estimator—the estimator induced by the oracle design that knows the ICC—decreases linearly with the budget B, i.e. it is $O(1/B)$. I then show that the simple `EtC-CS` algorithm induces a negligible excess variance on the estimator when compared to the optimal policy. In particular, the excess variance of `EtC-CS` is $O(\ln B/B^2)$. I further show that its excess variance is minmax optimal for large B by deriving matching lower bounds (up to logarithmic terms) of $O(1/B^2)$. Finally, I argue that this excess variance is non-trivial, at least when compared to canonical static policies. For reference, any static policy that takes a misspecified correlation as an input (including the minmax optimal static policy) will incur excess variance of $O(1/B)$, that is of the same order than the oracle variance and much larger than the excess variance of `EtC-CS`.

Alongside its performance guarantees, the main strength of `EtC-CS` is the simplicity and implementability of its two-stage design. However, one may wonder what would happen if the designer could implement a more adaptive sampling protocol. Consider the following setting. Given a compact prior for the ICC, the optimal number of clusters for an experiment may range between 20 and 60 clusters. According to `EtC-CS`, the researcher would sample a few units from 20 clusters, and then estimate the ICC coefficient. Imagine, that the estimated ICC prescribed implementing a 50 cluster design. Naturally, the researcher may need not to commit to the 50 cluster design straightaway. For instance, she could *safely* explore 10 more clusters to improve ICC estimation with minimal risk of incurring cluster oversampling. Based on this heuristic, we propose an improved algorithm where the designer samples clusters sequentially until she stops according to a data-dependent rule and commits to a final design. We refer to this refined adaptive design as Adaptive Cluster Selection (`ACS`). We show that `ACS` is also near-optimal with excess variance of $O(\ln B/B^2)$. The main challenge in the analysis of `ACS` compared to the analysis of `EtC-CS` is obtaining estimator guarantees under an adaptive stopping protocol. We tackle this problem in two ways: through time-uniform bounds for hard budgets, and through anytime bounds for unbounded budgets. A large array of simulations show the improved performance of `ACS` compared to `EtC-CS`, as well as the improved performance of both methods compared to static policies.

Finally, the reader may fear that adaptive design selection compromises inference. After all, data collection depends on pilot data, thus design-conditional inference cannot be taken for granted

(Wager (2024)). We use a very simple intuition: sample averages are asymptotically independent of sample variances under an error symmetry assumption. We then use this fact to prove a very strong result of independent interest for the literature on inference with adaptive data collection: inference on sample averages is invariant to adaptive data collection procedures depending on sample variances. This means that asymptotic normality of sample averages is preserved without further correction nor data wastage. The adaptive designs in this paper depend on (sequences of) ICC estimators, which are functions of within-cluster and between-cluster variances, therefore the theory applies. Simulation evidence confirms the theoretical prescriptions.

Empirical Application - TBC

Extensions - TBC

The rest of the manuscript is organized as follows, Section 2 reviews the literature in survey theory and adaptive experimental design, Section 3 introduces the cost-aware variance minimization problem under known ICC, Section 4 introduces the two candidate algorithms `EtC-CS` and `ACS`, and provides upper bound theoretical guarantees on excess variance, Section 5 derives a matching lower bound on the problem and compares the algorithms to prevalent static policies, Section 6 discusses inference under adaptive designs and particularizes the results to our newly derived algorithms, Section 7 collects simulation evidence confirming our theoretical guarantees, and Section 8 concludes.

2 Literature Review

Survey Theory. The problem of optimal multi-stage survey design under a budget is a well-studied problem in the field of survey theory dating back to the seminal work of Kish (1965).[5] On their simplest form, multi-stage surveys are deployed by first splitting the population of interest in a set of (non-overlapping) clusters, then, randomly selecting a number of clusters K (proportionally to their size), and, finally, randomly sampling N units from each of the selected clusters. The process may be towered by first drawing K_1 super-clusters, then K_2 sub-clusters, etc. The goal in this literature is very close to the spirit of this paper: to minimize the variance of the resulting population estimators by careful definition and/or selection of clusters and surveyed units. The literature can be divided in three broad categories: (i) sampling designs that take the clusters as given and optimize for unit selection, (ii) sampling designs that optimize cluster definition, and (iii) sampling designs that jointly optimize for cluster definition and unit selection (Ballin and Barcaroli (2013)). Our manuscript does take cluster partitions as given, thus we discuss the first category in further detail.

Stratified Survey Design. This sub-literature finds its most prominent representative on Bethel's algorithm (Bethel (1989)), an extension of Neyman's allocation to budget constraints and generic convex costs (Tschprow (1923), Neyman (1992)). Since the popularization of Bethel's algorithm, the literature has focused on extending its logic to multi-domain, multi-variable and multi-objective settings, with a particular focus on algorithmic performance (Barcaroli et al. (2022)). Surprisingly, despite (i) the field's impressive technical machinery and (ii) its emphasis on the role of intra-cluster correlation for optimal design,[6] to the author's knowledge there has been no attempt at introducing

[5] see Biemer and Lyberg (2003) for a more recent textbook treatment, and Sedgwick (2015) for a brief discussion and policy guide.

[6] See for instance Cicchitelli et al. (1992) for an extended discussion on the role of ICC and the inclusion of several nested clusters. Knox and Chondros (2004) presents compelling empirical evidence on the variability of the ICC across

ICC estimation as part of the design optimization problem. Recent treatments as Barcaroli et al. (2022) assume that the cluster correlation can always be estimated from auxiliary data. In practice, the design selection seems driven by *feel*, broad logistic considerations and common misconceptions. To name a few, the UN Handbook of Household Sample Surveys in Developing and Transition Economies Tam (2005) recommends to always sample as many clusters as logistically feasible. Other sources like Sedgwick (2015) argue that the number of clusters must be fixed in advance. I will argue that none of these considerations are necessarily optimal.

The Econometrics of Stratified Experiments. My work is closely related in spirit to the econometric literature on optimal stratification and optimal covariate adjustment, albeit the designs in this literature are not really adaptive. The goal of this literature is to minimize the variance of an ATE estimator. Some of it even considers the role of sampling costs. Some examples include Cytrynbaum (2021), Cytrynbaum (2024), Bai (2022) or Bai et al. (2024). Despite being closely related theme-wise, this paper does not look at the problem of strata definition nor covariate adjustment, but at the trade-off between cluster and unit selection under a budget. Moreover, on its simplest form, the proposed solution does not rely on external covariates for its implementation.

Adaptive Experimental Design. Formally, my work resonates with the adaptive experimental design and online policy learning literature. For instance, in the context of best arm identification, Adusumilli (2026) considers the problem of optimal stopping in sampling experiments, Imbens et al. (2025) shows the inadmissibility of uniform sampling rules in multi-arm settings, and Kasy and Sautmann (2021) discuss sequential treatment allocation. We further encounter theoretical and applied papers on bandit learning in Economics with a similar notion of regret to the one described in this paper such as Cesa-Bianchi et al. (2025) and Li et al. (2026).

Inference on Adaptive Designs. The literature on inference with adaptive designs is still quite thin, most coming from inference in the bandit literature. Results in this literature are overall quite pessimistic. See Nie et al. (2018) where the authors argue that most adaptive data collection procedures deliver negatively biased sample average estimates. Similarly, Hadad et al. (2021) show that IPW estimators are not asymptotically normal and suffer from fat tails.[7] Unlike most data collection procedures in this literature, our algorithms do not depend on the levels but on the second moments of the data. This difference allows us to escape from the malaise of adaptive data collection (at least under homogeneous clusters with balanced designs). Asymptotic independence is established under an error symmetry assumption, which resembles results in randomization inference as the ones in Canay et al. (2017).

Others. Finally, a shallow connection exists between my work and the bandit literature on optimal sampling designs, going back to the equivalence result of D-optimal and G-optimal problems in Kiefer and Wolfowitz (1960).[8] Nonetheless, the tools and scope in this literature are quite removed from the analysis in this manuscript.

domains. Campbell et al. (2004) collect a methodological note on the usage and reporting of ICC estimates.

[7]See Wager (2024) for a recent textbook treatment.

[8]See Lattimore and Szepesvári (2020) for a more recent textbook treatment.

3 Design Problem under Known ICC

3.1 Survey Design Problem

Consider a population with $\overline{K}$ equally sized clusters containing $\overline{N}$ units each. The goal of the researcher is to minimize the MSE of an estimator of a population average by deciding how many units $N_k \ll \overline{N}$ sample for each cluster, where $K = \sum \mathbb{1}(N_k > 0) \ll K$ is the number of sampled clusters. Clusters are indexed by $k = 1, \ldots, K$ and units are indexed by $i = 1, \ldots, N_k$ for each cluster k. Let y_{ki} be the variable of interest which follows an additive model with nested errors,[9]

$$y_{ki} = \mu + u_k + \varepsilon_{ki}, \tag{1}$$

where y_{ki} is the variable of interest, u_k is the cluster-level error (*iid* with mean 0 and variance ρ), ε_{ik} is the individual-level error (also *iid* with mean 0 and variance σ), and $u_k \perp\!\!\!\perp \varepsilon_{ki}$ for all k, i. Define $\lambda = \sigma/\rho$ as the relative **intra-cluster correlation** (ICC). Observe that the model can be re-parameterized under $\omega = (\mu, \rho, \lambda)$. We further assume that

Assumption 3.1. *Compact Parameter Space. The parameters of the model ω live in a known compact set $\Omega = [\mu_m, \mu_M] \times [\rho_m, \rho_M] \times [\lambda_m, \lambda_M]$ where $[\rho_m, \rho_M]$ is a strict subset of $(0, \infty)$.*

The model in Equation 1 implies the following moment restrictions

i) cross-cluster independence, i.e. $\text{CoVar}(y_{ki}, y_{lj}) = 0$ for all i, j and $k \neq j$,

ii) homogeneous total variance, i.e. $\text{Var}(y_{ki}) = \rho(1 + \lambda)$ for all k, i,

iii) within-cluster equicorrelation, i.e. $\text{CoVar}(y_{ki}, y_{kj}) = \rho$ for all i, j, and all k,

It follows that $\lambda = \text{Var}(y_{ki})/\text{Covar}(y_{ki}, y_{kj}) - 1$—what explains why it is referred to as (relative) intra-cluster correlation.[10] For simplicity in the exposition, we focus on the case where $N_k \in \{0, N\}$ for all k. In the homogeneous environment setting this is without loss, as the variance minimizing allocation is indeed a balanced one.[11] The general setting is discussed in Appendix A. For fixed (K, N), the natural estimator of the population parameter μ is simply the **average of averages**,

$$\hat{\mu} = \hat{\mu}(K, N) = \frac{1}{KN} \sum_{k=1}^{K} \sum_{i=1}^{N} y_{ki}$$

which is unbiased for μ under multi-stage randomization (and mean-zero noise).

3.2 Experimental Design

The same design problem arises in a simple cluster-stratified experiment. Suppose the researcher samples an even number of clusters $K \ll \overline{K}$ with $N \ll \overline{N}$ units per cluster. She then randomly assigns

[9] The variance minimization problem and excess variance analysis only rely on certain moment restrictions implied by Equation 1. In that sense, our analysis extends beyond this additive model. However, the additive error structure of the model plays an important role in our inference analysis, thus, for internal consistency and simplicity, we take Equation 1 as the model of reference.

[10] The literature defines the ICC in slightly different ways including ρ/σ or $\rho/(\rho + \sigma)$. Our results are invariant to the ICC definition under consideration, beyond the restriction $\rho > 0$ to avoid singularity.

[11] The lower bounds in Section 5 do not restrict to policies that only sample balanced allocations. Therefore, our balanced allocation algorithms are near-optimal in a very general sense.

treatment at the cluster level to $K/2$ clusters. Let $D_k \in \{0, 1\}$ denote the treatment status of cluster k. Using standard potential-outcomes notation, let $Y_{ki}(1)$ and $Y_{ki}(0)$ denote the treated and untreated potential outcomes of unit i in cluster k, and define the observed outcome by

$$Y_{ki} = D_{ki}Y_{ki}(1) + (1 - D_{ki})Y_{ki}(0)$$

The object of interest is the average treatment effect

$$\text{ATE} = \mathbb{E}[Y_{ki}(1) - Y_{ki}(0)] = \mu_1 - \mu_0$$

To parallel the survey framework, assume that each potential-outcome process follows the same nested-error structure as in Equation 1, i.e. $Y_{ki}(d) = \mu_d + u_k^d + \varepsilon_{ki}^d$ for $d \in \{0, 1\}$, with $u_k^d \perp\!\!\!\perp \varepsilon_{ki}^d$.[12] In this case, the model is parameterized by $\omega = (\text{ATE}, \mu_0, \rho, \lambda)$. Given the balanced randomization within each sampled cluster, the natural estimator of the average treatment effect is the average of within-cluster differences in means:

$$\hat{\text{ATE}}(K, N) = \frac{2}{K}\left(\sum_{k:D_k=1}^{K} \frac{1}{N}\sum_{i=1}^{N} y_{ki} - \sum_{k:D_k=0}^{K} \frac{1}{N}\sum_{i=1}^{N} y_{ki}\right)$$

Under multi-stage randomization, mean-zero noise and independence of potential outcomes, $\hat{\text{ATE}}(K, N)$ is unbiased for the ATE.

3.3 Variance Minimization under Linear Costs

The problem is completely symmetric in the survey and experimental design settings, so we focus on the survey setting as the operating framework. In both settings, the goal of the learner is to minimize the MSE of the estimator $\hat{\mu}$ subject to linear costs and a hard budget B. Under two-stage randomization, $\hat{\mu}$ is unbiased for μ, hence, the problem reduces to a constrained variance minimization problem

$$\min_{K\geq 2, N\geq 2} W(K, N, \omega) = \frac{1}{K}\rho\left(1 + \frac{\lambda}{N}\right) \qquad \text{s.t.}\;\; FK + VKN \leq B \tag{2}$$

where $W(K, N, \omega) = \text{Var}(\hat{\mu}(K, N))$, F is the fixed cost of sampling an additional cluster, V is the variable cost of sampling an additional unit and $B \geq 2(F + 2V)$ is the hard budget.[13] The equality follows from simple algebra under cross-cluster independence. The variance for $\hat{\text{ATE}}$ in the experimental design context is exactly the same. From now on, we leave integer considerations aside and focus on the **continuous relaxation** of the problem in Equation 2.[14]

A first immediate result is that for any $\rho > 0$, $\hat{\mu}(K, N)$ is consistent for μ if and only if $K \to \infty$. Intuitively, large N can only reduce the individual level noise ε_{ik}, while large K is necessary

[12] Clearly, independence may not hold across treatment status, i.e. $(u_k^0, \varepsilon_{ki}^0) \not\perp\!\!\!\perp (u_k^1, \varepsilon_{ki}^1)$.

[13] The lower bound $K \geq 2, N \geq 2$ has been imposed for technical reasons that affect the estimation of the parameters in Section 4. In practice, this restriction is with little loss as *all* clustered surveys/experiments consider larger samples anyway.

[14] Although the theory is derived by disregarding integer considerations, effective algorithm implementation is integer constrained. All throughout the text, including Tables and Graphs, final integer allocation has been produced by considering the two nearest integer neighbors to the continuous solution N^*, adjusting K^* accordingly and selecting the variance minimizer between the two candidates subject to budget feasibility. This procedure has been applied to own and competitor algorithms to level the field and ease cross-algorithms comparison. Replicating code can be found in the Supplementary Material.

to drive the the cluster level noise u_k to 0. Finally, given the linearity of the cost structure, the problem admits a representation as an *unconstrained* minimization problem over a single design variable N. Solving from the budget constraint

$$K^*(N) = \frac{B}{F+VN}, \qquad \min_{2\leq N\leq (B-2F)/(2V)} \frac{F+VN}{B}\rho \cdot \left(1+\frac{\lambda}{N}\right)$$

We can solve for the optimal design $(K^*(\omega), N^*(\omega))$ by using first order conditions, i.e.

$$N^*(\omega) = \min\left\{\max\left\{2, \sqrt{\lambda\frac{F}{V}}\right\}, \frac{B-2F}{2V}\right\}, \qquad K^*(\omega) = \frac{B}{F+VN^*(\omega)} \tag{3}$$

We note immediately that the map $(N^*(\omega), K^*(\omega))$ depends on ω only through the relative within-cluster correlation λ (and not the location nor scale parameters). Accordingly, define the **optimal map** as $\lambda \mapsto (N^*(\lambda), K^*(\lambda))$. Despite its name, the optimal map is only really *optimal* if the designer takes the true λ as input. In general, applying the optimal map to a misspecified λ can have sizable implications on the design and the variance of the estimator induced by this design. Figure 1 plots $(N^*(\lambda),\ K^*(\lambda))$ and the excess variance of the resulting estimator given the design $(N = N^*(\lambda), K = K^*(\lambda))$ for different values of λ compared to the oracle policy $(N = N^*(\lambda_0), K = K^*(\lambda_0))$. Figure 1 confirms that small λ mispecification can have large implications on the induced design and the resulting excess variance.

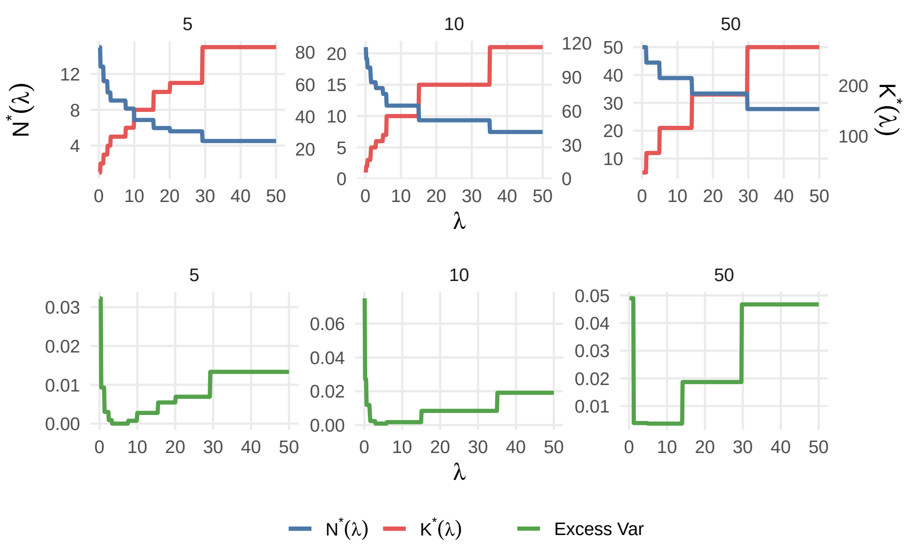

Figure 1: $(N^*(\lambda), K^*(\lambda))$ (left) and excess variance $W(K^*(\lambda), N^*(\lambda), \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0)$ (right) for different F/V ratios. Update values $B = 1000, V = 2,\ \omega = (0, 1, 5)$

The variance of the oracle policy that has access to true $\lambda = \lambda_0$ is just

$$W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) = \frac{\rho_0\left(\sqrt{V\lambda_0} + \sqrt{F}\right)^2}{B}$$

Any sampling policy which does not know λ_0 will find in the **oracle variance** a natural benchmark to compete against. Finally, to avoid dealing with corner solutions, we include an assumption to guarantee interior solutions of the continuous relaxation problem in Equation 2.

Assumption 3.2. *Interior Solution. Given a compact set Ω such that $\lambda \in [\lambda_m, \lambda_M]$, $N^*(\lambda_m) \geq 2$ and $N^*(\lambda_M) \leq \frac{B-2F}{2V}$, with $B \geq 2(F + 2V)$.*

Assumption 3.2 can also be interpreted as the parameter space being compatible with the implementation of a non-degenerate cluster experiment.[15]

4 Design Problem under Unknown ICC

4.1 Estimation of λ

Optimal survey design under unknown model parameters will depend on some form of parameter estimation, therefore, we start this section by characterizing the relevant estimators. Fix (K, N) with $K \geq 2, N \geq 2$ and define

$$\hat{\mu}^k(N) = \frac{1}{N}\sum_{i=1}^{N} y_{ki} \qquad \tilde{w}(K,N) = \frac{1}{K(N-1)}\sum_{k=1}^{K}\sum_{i=1}^{N}(y_{ki} - \hat{\mu}^k(N))^2$$
$$\tilde{b}(K,N) = \frac{1}{K-1}\sum_{k=1}^{K}(\hat{\mu}^k(N) - \hat{\mu}(K,N))^2 \qquad \tilde{\rho}(K,N) = \tilde{b}_k - \frac{\tilde{w}_k}{N}$$

where $\hat{\mu}^k$ is the sample average for cluster k, $\tilde{w}$ is the within-cluster sample variance, $\tilde{b}$ is the between-cluster sample variance and $\tilde{\rho}$ is sample estimator for ρ. Now define the clip operator $\Pi_{[a,b]}(x) = \min\{b, \max\{a, x\}\}$ and consider the clipped analogues

$$\hat{w}(K,N) = \Pi_{[\lambda_m\rho_m, \lambda_M\rho_M]}(\tilde{w}(K,N)) \qquad \hat{\rho}(K,N) = \Pi_{[\rho_m,\rho_M]}(\tilde{\rho}(K,N)) \qquad \hat{\lambda}(K,N) = \frac{\hat{w}(K,N)}{\hat{\rho}(K,N)}$$

where $\hat{\lambda}(K,N)$ is the (clipped) sample estimator of λ. Lemma 4.1 summarizes the properties of these estimators

Lemma 4.1. *Estimator Properties. Let $\xrightarrow[T]{p}$ note convergence in probability under T asymptotics, then*

1. *$\hat{w}(K,N) \xrightarrow[N]{p} \sigma$ (with fixed $K \geq 1$) and $\hat{w}(K,N) \xrightarrow[K]{p} \sigma$ (with fixed $N \geq 2$),*
2. *$\hat{\rho}(K,N) \xrightarrow[K]{p} \rho$ (with fixed $N \geq 2$),*
3. *$\hat{\lambda}(K,N) \xrightarrow[K]{p} \lambda$ (with fixed $N \geq 2$)*

There are two important takeaways from the definitions in Equation 4.1 and Lemma 4.1. Firstly, the ICC is essentially the ratio of the within-cluster to the between cluster variance, and, as such, it can be consistently estimated as the ratio of the sample analogues of these two objects. Secondly, while the within-variance can be estimated consistently under both N and K asymptotics,

[15] Assumption 3.2 uses the fact that N^* is increasing in λ. Moreover, the condition $N^*(\lambda_M) \leq (B - 2F)/2V$ can be equally stated as $K^*(\lambda_M) \geq 2$. The assumption can then be further interpreted as the minimal clustered experiment $N = 2, K = 2$ being *worst-case* implementable. See Section 4 for further discussion.

the between-variance (and therefore ρ) can only be consistently estimated under K asymptotics. As a result, only K asymptotics deliver consistent estimation of λ.

This fact reveals an important learning heuristic that will play a central role in the derivation of the adaptive algorithm. Imagine that the designer uses part of her resources to learn λ by sampling some pilot data (K_p, N_p), so she can eventually implement the near-optimal mapping $(K^*(\hat{\lambda}(K_p, N_p)), N^*(\hat{\lambda}(K_p, N_p)))$ in the second stage. The designer will find that using most of her *exploration budget* for cluster exploration—small N_p and large K_p—is close-to-optimal for λ estimation.[16]

4.2 Explore-then-Commit Cluster Sampling

In this section we propose a simple solution to the problem of the designer under unknown ICC using a Explore-then-Commit structure. Imagine that the learner conducts a small pilot to estimate λ by sampling K_p clusters and N_p units per cluster. She can then construct an estimator $\hat{\lambda}_p = \hat{\lambda}(K_p, N_p)$ according to Equation 4.1 and recover the predicted optimal map $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$. Now, observe that predicted optimal map will be (budget) implementable iff $K_p \leq K^*(\hat{\lambda}_p)$ and $N_p \leq N^*(\hat{\lambda}_p)$.

Under the interior solution Assumption 3.2 (and the clipping of the estimators), a safe way to guarantee the implementability of the predicted optimal map is setting $K_p = K^*(\lambda_M)$ and $N^*(\lambda_m)$. Then, for any $\hat{\lambda}_p$, it holds that $K_p \leq K^*(\hat{\lambda}_p)$ and $N_p \leq N^*(\hat{\lambda}_p)$.[17] We refer to this two-stage protocol as Explore-then-Commit Cluster Selection (`EtC-CS`). Algorithm 1 formalizes its structure and Theorem 4.3 provides upper bound guarantees on its performance under a semi-parametric restriction on the error terms u_k, ε_{ki}.

Algorithm 1 `EtC-CS`

Input $B, F, V, \lambda_m, \lambda_M$
Sample $K_p = K^*(\lambda_M)$ clusters and $N_p = N^*(\lambda_m)$ units per cluster
Recover $\hat{\lambda}_p = \hat{\lambda}(K_p, N_p)$
Implement $K^*(\hat{\lambda}_p)$ and $N^*(\hat{\lambda}_p)$

For any policy π, define the excess variance of the policy as the random variable governing the difference between the variance of the design (K_π, N_π) induced by policy π and the variance of the oracle policy, i.e.

$$\Delta_\pi = W(K_\pi, N_\pi, \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0)$$

Assumption 4.2. *Errors Subgaussianity Let u_k be $\sqrt{\kappa_u}$-subgaussian and ε_{ki} be $\sqrt{\kappa_\varepsilon}$-subgaussian*

The subgaussian assumption is quite standard in the online learning and policy learning literature. It essentially enforces sufficient concentration of the error terms. Similar derivations will go through under a weaker subexponential assumption.

[16] Formally, one can show that the N-minimizer of the asymptotic variance of any regular estimator $\hat{\lambda}$ is finite and independent of the budget, while the K-minimizer is nearly linear in budget. This justifies treating N as fixed and increasing K for λ estimation.

[17] Observe that, in some sense, the proposed design is minmax optimal for a learner that wants to implement a second stage optimal mapping of the form $(K^*(\hat{\lambda}), N^*(\hat{\lambda}))$.

Theorem 4.3. *Upper bound on `EtC-CS`. Under Assumption 3.2 and Assumption 4.2, with probability at least $1-\alpha$, there exists a constant $C < \infty$ independent of B such that*

$$\Delta_{\texttt{EtC-CS}} \leq C \; \frac{\ln(1/\alpha)}{B^2}$$

Interpretation of Theorem 4.3. Proof of Theorem 4.3 can be found in Appendix B. From our discussion above, `EtC-CS` does implement a near optimal design given by $K^*(\hat{\lambda}_p)$ and $N^*(\hat{\lambda}_p)$. Therefore, deviations with respect to the oracle variance are local to λ, thus of second order. Because the effective variance of $\hat{\lambda}_p$ is the cluster size of the pilot K_p, and $K_p = K^*(\lambda_M) = B/(F + VN^*(\lambda_M))$ is linear in budget, the overall excess variance is $O(1/B^2)$. Theorem 4.3 shows that, with high probability, the excess variance of `EtC-CS` of $O(1/B^2)$ is asymptotically negligible compared to the variance of the oracle policy of $O(1/B)$. It then follows that a learner which explores according to `EtC-CS` is (asymptotically) not worse than an oracle designer who knows λ ex-ante.

4.3 Adaptive Cluster Selection

While theorists emphasize the superior performance of adaptive designs, practitioners often encounter logistical challenges that prevent their full implementation (Chow and Corey (2011)). In that sense, we praise the efficacy and simplicity of `EtC-CS` which achieves near-optimal performance guarantees with minimal adaptability—just a two-stage design. However, one could wonder what can be achieved if the designer was given access to further adaptability. Consider the following toy scenario. A researcher must select a design that samples something between 20 and 60 clusters, i.e. given a fixed budget $K^*(\lambda_M) = 20$ and $K^*(\lambda_m) = 60$. `EtC-CS` would then prescribe to sample 20 clusters, compute $\hat{\lambda}_p = \hat{\lambda}(20, N_p)$, and implement $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$. Now imagine that the design prescribed by the pilot would recommend sampling $K^*(\hat{\lambda}_p) = 50$ clusters. It would seem unreasonable to commit straightaway to this 50 cluster design, especially if the learner was able to reject the null $H_0 : K^*(\lambda_0) < 40$ at standard confidence levels. In practice, subject to implementability, the designer could safely explore a few more clusters to get a better sense of the optimal design with minimal risk of cluster oversampling—the biggest concern behind the minmax pilot design in `EtC-CS`. Based on this heuristic, we propose a more flexible algorithm where the designer samples clusters sequentially until she stops according to a data-dependent rule and implements the final design. We refer to this algorithm as Adaptive Cluster Selection (`ACS`). In particular, consider an algorithm that sequentially explores clusters until the stopping time τ, defined by

$$\tau = \inf\{K : K + 1 > K^*(\hat{\lambda}_K)\} \wedge \left\lfloor \frac{B}{F + VN_p} \right\rfloor \tag{4}$$

where $\hat{\lambda}_K = \hat{\lambda}(K, N_p)$. In other words, the adaptive algorithm explores N_p units from an additional cluster until sampling the new cluster would incur oversampling according to the online estimator of optimal design. By treating N_p as fixed, and exploring only in the K dimension, this heuristic sampling process combines simplicity with statistical guarantees. In principle, under a similar heuristic, the learner may consider sampling a new unit—either from an already explored cluster or from a new cluster—to maximize the (cost adjusted) expected marginal return to exploration. However, this allocation process is not only computationally infeasible but also highly impractical in real applications.

In addition, as per the discussion earlier in this Section, reducing the exploration to the K dimension is the correct simplification. In small samples, sampling additional units from already explored clusters may reduce the variance of the $\hat{\lambda}$ estimator. However, the asymptotic variance is independent of N and only strictly decreasing in K. It follows that, when it comes to λ estimation, asymptotically, the learner has an incentive to commit all her exploration resources to cluster exploration. Algorithm 2 formalizes this heuristic procedure and Theorem 4.4 provides upper bound guarantees on its performance.

Algorithm 2 `ACS`

> **Input** $B, F, V, \lambda_m, \lambda_M$
> **Initialize** $K = 0$, $N_p = N^*(\lambda_m)$, $\hat{\lambda}_0 = \lambda_m$
> **while** $K + 1 \leq \min\left\{K^*(\hat{\lambda}_K), \left\lfloor \frac{B}{F+VN_p} \right\rfloor\right\}$
>   **explore** an additional cluster k and **observe** $(y_{ki})_{i=1}^{N_p}$
>   **update** $K = K + 1$
> **recover** $\hat{\lambda}_{\texttt{ACS}} = \hat{\lambda}(K-1, N_p)$ using $((y_{ki})_{i=1}^{N_p})_{k=1}^{K-1}$
> **implement** $K^*(\hat{\lambda}_{\texttt{ACS}})$ and $N^*(\hat{\lambda}_{\texttt{ACS}})$

Beyond adaptability, the key distinction between Algorithms 1 and 2 is that `ACS` concentrates all cluster exploration during the pilot, while `EtC-CS` explores new clusters after the pilot. This allows `ACS` to use as many clusters as possible for λ estimation. Recall that the estimation of λ improves with K, which, incidentally, is the expensive element of the design. Hence, there is no reason to waste information for λ estimation by excluding clusters.[18]

The use of information from $K-1$ clusters for design estimation is a minor technical concern. Given a stopping time τ it must have been the case that $\tau \leq K^*(\hat{\lambda}_{\tau-1})$, otherwise the algorithm would have not explored cluster τ. Moreover, $\tau+1 > K^*(\hat{\lambda}_\tau)$. However, from these two facts alone one cannot infer the natural condition $\tau \leq K^*(\hat{\lambda}_\tau)$, but only $\tau \leq K^*(\hat{\lambda}_{\tau-1})$. Once again, deviations with respect to the oracle variance are local to λ, hence of second order, but with improved performance guarantees on $\hat{\lambda}_{\tau-1}$ compared to $\hat{\lambda}_p$. Nonetheless, from a technical perspective, it is difficult to translate this intuitive improvement into an analytical improvement. To see why, observe that the learner needs to control deviations of the form $\mid \hat{\lambda}_{\tau-1} - \lambda_0 \mid$ where τ is a random stopping time governed by a complicated law. We circumvent this issue by deriving uniform bounds over the entire sequence of possible stopping times $\tau \in [K^*(\lambda_M), K^*(\lambda_m)]$. Mathematically, we derive bounds $D_K^\lambda(\alpha)$ such that

$$\mathbb{P}(\mid \hat{\lambda}_K - \lambda_0 \mid \leq D_K^\lambda(\alpha) \ \forall \ K \in [K^*(\lambda_M), K^*(\lambda_m)]) \geq 1 - \alpha$$

provided $\mathbb{P}_{\texttt{ACS}}(\mid \hat{\lambda}_{\tau-1} - \lambda_0 \mid \leq D_\tau^\lambda(\alpha)) \geq \mathbb{P}(\mid \hat{\lambda}_K - \lambda_0 \mid \leq D_K^\lambda(\alpha) \ \forall \ K \in [K^*(\lambda_M), K^*(\lambda_m)])$, where the first law is given by the interaction of the environment and the algorithm (while the second only depends on the randomness of the environment). We make two important remarks on this regard

1. This bound relies explicitly on the maximum number of clusters that the learner could possibly explore, i.e. $K_m = K^*(\lambda_m)$, through a union bound argument. When K_m is very big, the

[18] This dynamic can also be modeled in terms of incentives between a λ learner and a maximally risk averse designer that wants to implement an optimal map $(K^*(\hat{\lambda}), N^*(\hat{\lambda}))$ at all costs. During the first $K^*(\lambda_0)$ clusters, the preferences of both individuals is perfectly aligned.

designer could instead consider anytime bounds which guarantee coverage for the infinite sequence $\mathbb{P}(|\ \hat{\lambda}_K - \lambda_0\ | \leq D_K^{\lambda}(\alpha)\ \forall\ K \geq K^*(\lambda_M))$. These bounds rely on the martingale properties of mixing likelihood-ratios as in Howard et al. (2021). This anytime procedure delivers bounds of the same order but tighter in some contexts at the expense of minor additional assumptions. Moreover, anytime bounds are more amenable to Bayesian analysis. As such, they are discussed extensively in the Supplementary Material.

2. The uniform bound coverage does not come for free and induces an additional $O(\ln B)$ excess variance compared to the excess variance of `EtC-CS`. Intuitively, `ACS` should do no worse than `EtC-CS`, what suggests that the additional $\ln B$ term is a byproduct of the analysis and not a decrease in performance. The seemingly uniform dominance of `ACS` over `EtC-CS` is confirmed via simulation evidence.[19]

Theorem 4.4. *Upper bound on `ACS`. Under Assumption 3.2 and Assumption 4.2, with probability at least $1-\alpha$, there exists a constant $C < \infty$ independent of B such that*

$$\Delta_{ACS} \leq C\ \frac{\ln(B/\alpha)}{B^2}$$

Interpretation of Theorem 4.4. Theorem 4.4 recovers the same upper bounds on excess variance (up to a logarithmic term) than `EtC-CS`. As a result, `ACS` remains asymptotically not worse than the oracle designer. Moreover, the simulation evidence in Section 7 suggests a non-negligible improvement in performance coming from the additional design's flexibility.

4.4 Worst Case and Expected Excess Variance

In this section we translate our high probability upper bounds on excess variance into upper bounds on expected excess variance. To do so, we first characterize the worst case excess variance of any design which consumes the budget entirely. We then argue that with probability α the designer incurs worst case excess variance, and with probability $1-\alpha$ excess variance of order $O(1/B^2)$ and $O(\ln B/B^2)$ as discussed in Theorems 4.3 and 4.4. In both cases, we show upper bounds on expected excess variance of $O(\ln B/B^2)$, preserving the asymptotically negligible excess variance property of the high probability bounds.

Lemma 4.5. *Worst Case Excess Variance. For any policy π such that $FK_\pi + VK_\pi N_\pi = B$ with $K_\pi \geq 2, N_\pi \geq 2$ and sufficiently large B, there exists a small positive δ such that*

$$W(K_\pi, N_\pi, \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) \leq \frac{\rho_M}{2} + \delta\ \ \forall\ \omega_0 \in \Omega$$

Interpretation of Lemma 4.5. As discussed in the problem's introduction, estimator's variance is only effectively driven to 0 under K asymptotics. As a result, any design which samples a finite number of clusters will incur non-vanishing excess variance. In particular, the lower bound in

[19] Our analysis leverages standard uniform bounds from the online learning literature. Improved bounds compared to the ones used in our analysis have the ability to remove the logarithmic term entirely at the expense of a more cumbersome analysis, see `MOSS` style bounds in Lattimore and Szepesvári (2020). The simulation evidence and strong intuition that `ACS` should uniformly outperform `EtC-CS` moved us to consider the *easy* bounds with the additional $\ln(B)$ term in the numerator. Anyway, this will not matter when considering upper bounds on expected excess variance, and not just high-probability coverage, as in that regime, both `EtC-CS` and `ACS` display excess variance of $O(\ln B/B^2)$.

excess variance of these policies is $\rho_0/2$. The usual compactness assumption on Ω recovers the final expression.

Theorem 4.6. *Expected Excess Variance. Under Assumption 3.2, for sufficiently large B, there exists constants $C_1, C_2 < \infty$ independent of B such that*

$$\mathbb{E}_{\mathit{EtC\text{-}CS}}\left[W(K,N,\omega_0)\right] - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) \le C_1 \frac{\ln(B)}{B^2}$$
$$\mathbb{E}_{\mathit{ACS}}\left[W(K,N,\omega_0)\right] - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) \le C_2 \frac{\ln(B)}{B^2}$$

Interpretation of Theorem 4.6. Controlling for expected excess variance is a stronger performance guarantee than controlling for excess variance with high probability. The learner can rest assured that switching across metrics does not worsen the order of excess variance beyond a $\ln(B)$ term. As a result, the excess variance of both `EtC-CS` and `ACS` remains asymptotically negligible compared to the oracle with an upper bound of $O(\ln B/B^2)$. In the next Section, we argue that the rate in the excess variance of these algorithms is not only negligible but also near-optimal—up to logarithmic terms. This means that no algorithm can substantially improve upon Algorithms 1 and 2. Finally, we show that their excess variance is also non-trivial, at least when compared to any misspecified static policy.

5 Lower Bound and Policy Comparison

In this Section we show that `EtC-CS` and `ACS` are essentially unimprovable by deriving lower bounds on the problem of $O(1/B^2)$, i.e. lower bounds are matching up to logarithmic terms. We then prove that any static policy which optimizes with respect to a misspecified ICC will incur excess variance of $O(1/B)$, that is, of the same order than the oracle variance and one order of magnitude larger than the excess variance of the proposed algorithms. We use this evidence to argue that the excess variance of our algorithms is indeed non-trivial and that ICC learning is necessary for good asymptotic performance in survey and experimental designs.

5.1 Lower Bound

As anticipated, we derive lower bounds for the generalized game where the learner can select N_k arbitrarily for every k (subject to budget constraints). See Appendix A for a complete characterization. In order to derive lower bounds for this regime, we consider an *easier* version of the problem where, every time the learner samples N_k units cluster from cluster k she does not only observe $(y_{ik})_{N_k}$ realizations, but she also observes the cluster noise u_k. The problem is easier inasmuch as it contains weakly more information for the learner than the standard experiment. By deriving lower bounds on the augmented experiment we effectively guarantee lower bounds on the benchmark game. This simplification saves us some technical problems. Because observations within the same cluster are correlated, when cluster k has already been visited, the new observations from that cluster are not identically distributed conditional on the history of observations. This lack of independence invalidates standard *divergence decomposition* arguments in the original experiment. By revealing the cluster noise u_k when cluster k is explored for the first time, history-conditional independence is restored and standard proof arguments go through. The rest of proof is quite mainstream. We consider two data generating processes y_-, y_+—indexed by ω_- and ω_+—following the structure of Equation 1. Define

$$\omega_- = (\mu, \rho_-, \lambda_-) \qquad \omega_+ = (\mu, \rho_+, \lambda_+) \qquad \lambda_- = \sigma/\rho_- \qquad \lambda_+ = \sigma/\rho_+$$

For some $\lambda \in (\lambda_m, \lambda_M)$ and some small positive δ, ρ_- and ρ_+ are selected such that $\lambda_- = \lambda - \delta$ and $\lambda_+ = \lambda + \delta$. In other words, the only difference across the DGPs is some small variation in the cluster variance ρ. As a result, optimizing with respect to λ_- is optimal in environment y_- and optimizing with respect to λ_+ is optimal in environment y_+. Figure 2 illustrates this intuition.

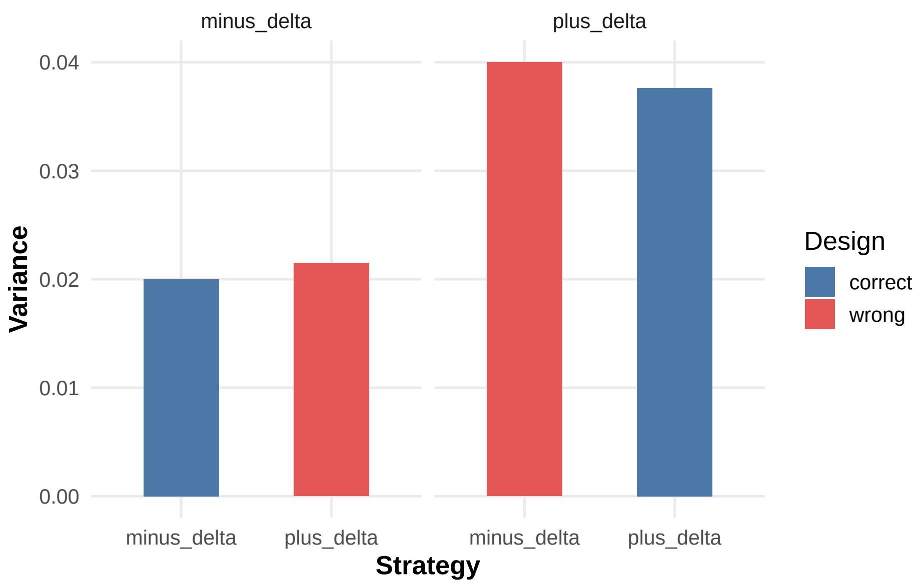

Figure 2: (True) variance $W(K, N, \cdot)$ under $\underline{\omega}$ (left) and $\overline{\omega}$ (right). $\rho = 1$, $\sigma = 5$, $\delta = 0.5$, $B = 2000$, $F = 20$, $V = 2$.

δ is selected to balance a dual-objective: punishment and identification. A large δ makes it easier for the learner to distinguish across the two environments (and, therefore, to optimize accordingly), but it also penalizes her more heavily in case she fails to identify the correct instance. Conversely, a smaller δ makes it more difficult to tell instances apart, but it also makes the additional excess variance of selecting a suboptimal design smaller. By selecting $\delta \sim O(B^{-1/2})$—as it is customary in learning problems—excess variance is maximized. Theorem 5.1 summarizes the main result in this Section. The proof of Theorem can be found in Appendix C.

Theorem 5.1. *Lower Bound on Excess Variance. For any policy $\pi \in \Pi$, there exists a DGP y and a positive finite constant $C < \infty$ such that*

$$\mathbb{E}_\pi \left[W(K, \boldsymbol{N}, \omega_0) \right] - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) \geq \frac{C}{B^2}$$

Observe that Theorem 5.1 establishes a lower bound for a very general policy class Π.[20] In particular, we allow policies to explore different number of units across different clusters, to revisit

[20]The exact definition of the policy class Π can be found in the Appendix. The restriction to the policy class Π essentially disregards policies which do not terminate in finite time almost surely or do not satisfy the budget constrain almost surely.

clusters freely, and to sample a different number of units every time the cluster is visited.

5.2 Policy Comparison

The results in the previous section suggest that our algorithms are essentially unimprovable. However, it is worth asking if the rate in the excess variance is actually trivial, in the sense that default policies would achieve a similar rate. Because this is the first paper that fits learning within survey design, we may only compare our algorithm to static design policies. In particular, we compare `EtC-CS` and `ACS` to any static policy that takes a misspecified parameter λ as input and optimizes over it, including the minmax optimal policy. By doing so, we show that the rate in the excess variance of Algorithms 1 and 2 is indeed non-trivial as all these policies accrue (expected) excess variance of $O(1/B) \gg O(\ln B/B^2)$.[21]

Lemma 5.2. *Misspecified Static Policies and Minmax Optimal Static Policy. Let $\lambda_b, \lambda_0 \in [\lambda_m, \lambda_M]$ such that $\lambda_b = \lambda_0 + b$ for some $b \in \mathbb{R}$. Define the static policy $\pi_b : (K_{\pi_b}, N_{\pi_b}) = (K^*(\lambda_b), N^*(\lambda_b))$, then for some positive finite constant C*

$$W(K_{\pi_b}, N_{\pi_b}, \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) = C\ \frac{b^2}{B} + O\left(\frac{b^3}{B}\right) \sim O(1/B)$$

Moreover, define the optimal minmax static policy $\pi_{mM} : (K_{\pi_{mM}}, N_{\pi_{mM}}) = (K^*(\lambda_{mM}, N^*(\lambda_{mM}))$ where

$$\lambda_{mM} = \arg\min_{\lambda} \max_{\omega\in\Omega} W(K^*(\lambda), N^*(\lambda), \omega) - W(K^*(\lambda), N^*(\lambda), \omega)$$

Then,

$$W(K_{\pi_{mM}}, N_{\pi_{mM}}, \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) = \frac{C(\lambda_{mM} - \lambda_0)^2}{B} + O\left(\frac{(\lambda_{mM} - \lambda_0)^3}{B}\right) \sim O(1/B)$$

The Lemma above shows that any misspecified static policy (including the minmax optimal) obtains a worst-case excess variance of the same order than the variance of the oracle policy itself and much larger than the worst-case excess variance of the learning protocols $O(\ln B/B^2)$. This result is not really surprising. When there is no scope for learning, any deviation from the true λ_0 leads to a fixed second-order deviation. This preliminary result suggest that the excess variance of the policy is indeed non-trivial, at least when compared against static designs.

6 Inference

So far, we have argued vehemently in favor of the adaptive design of cluster experiments. However, the experimentalist might be concerned that by selecting the design after looking at the pilot data, the distribution of the resulting estimator conditional on the design might be compromised. Under an error symmetry assumption, we argue that this is not the case and that the distribution of the resulting estimator remains invariant after conditioning on the designs induced by `EtC-CS` and `ACS`. This means that asymptotic normality of $\hat{\mu}(K, N)$ is preserved without further correction nor pilot

[21] Conditional on the design, the variance of the estimator is not random. As a result, all notions of variance (expected excess variance, excess variance in high probability, etc.) coincide for static policies that determine the exploration ex-ante.

data exclusion. This result is of independent interest to the literature on inference with adaptive data collection.

The proof can be split in 3 steps. In the first step we argue that the designs of interest are functions of (a sequence of) $\hat{\lambda}(K,N)$ estimators. We then argue that these estimators are asymptotically even. Recall that $\hat{\lambda}(K,N)$ is a function of the within-variance and the between-variance. The within-variance is even for all (K,N), but the between-variance is only asymptotically even, as it de-means by the grand-mean. In the second step, we define the average cluster-k errors $M_{k,N} = u_k + \frac{1}{N}\sum_{i=1}^{N}\varepsilon_{ki}$ (for some fixed N). We then leverage the cluster-symmetry assumption to show zero asymptotic covariance between $M_{k,N}$ and the sample variances through an odd-function even-event argument.

Assumption 6.1. *Cluster Symmetric Errors. Let $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ be the vector of cluster k errors as in Equation 1. Assume $C_k \overset{d}{=} -C_k$.*

This assumption is relatively common in randomization inference analyses as in Canay et al. (2017), yet, it remains strong. In particular, it rules out binary, count or skewed outcomes. The third step confronts the main technical challenge in the analysis. Intuitively, one would like to transform the weak convergence to independent limits of $M_{k,N}$ and the sample variances into convergence to independent limits of $\hat{\mu}(K,N) - \mu = \frac{1}{K}\sum_{k=1}^{K} M_{k,N}$ and the sample variances. The main problem is that the sample average function depends on the design, thus weak convergence to independent limits is not sufficient. Instead, one needs to show stable convergence. We do so by relying on a Martingale stable CLT invoking the martingale property of $M_{k,N}$. See Appendix D for further details.

Theorem 6.2. *Design Robust Inference Let (K_A, N_A) be the design induced by* `EtC-CS` *or* `ACS`*, with (K_A, N_A) $\mathcal{G}$-measurable. Define $\theta_0 = \rho_0(1 + \lambda_0/N^*(\lambda_0))$. Then, under Assumption D.1*

$$Z_B := \frac{\sqrt{K_A}\,(\hat{\mu}(K_A,N_A) - \mu)}{\theta_0} \to N(0,1) \quad \mathcal{G}\text{-stably}$$

Therefore, $\sup_x \mid \mathbb{P}(Z_B \leq x \mid (K_A,N_A)) - \Phi(x) \mid \xrightarrow{p} 0$. The same result holds when replacing θ_0 by $\hat{b}(K_A,N_A)$, thus

$$\hat{\mu}(K_A,N_A) \pm z_{1-\alpha/2}\sqrt{\hat{b}(K_A,N_A)/K_A}$$

has asymptotic coverage $1-\alpha$.

The complete set of regularity conditions (Assumption D.1), as well as the proof of Theorem 6.2 and the definition of the σ-algebra $\mathcal{G}$, can be found in Appendix D. The theorem conveys a very strong result. It claims that the distribution of the induced estimator remains invariant to adaptive data collection when the collection is driven by second moments of the data, i.e. it has the same asymptotic distribution as an estimator with a fixed design. The simulation evidence in Section 7 provides empirical support for these claims. Overall, this result opens the door for a generalized theory of inference under adaptive data collection when sampling is driven by estimates of second moments. Contrary to most results in the literature, it suggests that the researcher can preserve standard inference without further refinements nor data exclusion.

7 Simulations - TBC

8 Conclusion

This paper presents a first solution to the problem of clustered experimental and survey design under unknown intra-cluster correlation. In particular, it presents two algorithms with varying degrees of adaptability which turn-out to be minmax optimal in a setting with additive errors, linear costs and a fixed budget. The algorithms do not only display negligible excess variance compared to the oracle, but they also outperform significantly every misspecified static policy. In addition, under a symmetry assumption, we show that there is no trade-off between design optimality and inference, as the data collection protocols discussed in this manuscript do not interfere with asymptotic inference. Not without limitations, this is an encouraging result for adaptive data collection, and an attempt to push further adaptive experimentation and adaptive data collection.

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A Proofs Section 3

Generalized Problem. We first introduce the general problem where the learner can sample an arbitrary number of units N_k from each cluster k subject to budget constraints. Let $\boldsymbol{N}$ be the vector of non-empty samples, i.e. $\boldsymbol{N} = (N_1, \ldots, N_k, \ldots, N_K)$. In this setting, the estimator of interest is given by the average of averages

$$\hat{\mu}(K, \boldsymbol{N}) = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{N_k} \sum_{i=1}^{N_k} y_{ki}$$

where $K = \sum_k \mathbb{1}(N_k > 0)$. The variance of the estimator is given by

$$\begin{aligned} W(K, \boldsymbol{N}, \omega) &= \mathrm{Var}(\hat{\mu}(K, \boldsymbol{N})) = \mathrm{Var}\left(\frac{1}{K} \sum_{k=1}^{K} \frac{1}{N_k} \sum_{i=1}^{N_k} y_{ki}\right) = \mathrm{Var}\left(\mu + \frac{1}{K} \sum_{k=1}^{K} u_k + \frac{1}{K} \sum_{k=1}^{K} \frac{1}{N_k} \sum_{i=1}^{N_k} \varepsilon_{ki}\right) \\ &= \frac{K}{K^2}\rho + \frac{1}{K^2} \sum_{k=1}^{K} \frac{\sigma}{N_k} = \frac{\rho}{K}\left(1 + \frac{1}{K} \sum_{k=1}^{K} \frac{\lambda}{N_k}\right) = \frac{\rho}{K}\left(1 + \frac{\lambda}{N_H(\boldsymbol{N})}\right) \end{aligned}$$

where $N_H = \left(\frac{1}{K} \sum_{k=1}^{K} \frac{1}{N_k}\right)^{-1}$ is the harmonic mean. For $N_k = N$ the expression simplifies to Equation 2.

Variance Minimization. The generalized variance minimization problem in Equation 2 can be written as

$$\min_{K \geq 2,\ \boldsymbol{N}: N_{k \leq K} \geq 2} W(K, \boldsymbol{N}, \omega) \qquad \text{s.t. } KF + V \sum_{k=1}^{K} N_k \leq B$$

Optimal Map. Fix K and define $S = \sum_{k=1}^{K} N_k$. Minimizing the variance in terms of N_k reduces to minimize $\sum_{k=1}^{K} \frac{1}{N_k}$ subject to $S = \sum_{k=1}^{K} N_k$. The map $x \mapsto 1/x$ is strictly convex in $\mathbb{R}_+$, therefore (by Jensen's)

$$\frac{1}{K} \sum_{k=1}^{K} \frac{1}{N_k} \geq \frac{1}{\frac{1}{K} \sum_{k=1}^{K} \frac{1}{N_k}} = \frac{K}{S}$$

with equality iff $N_k = N = \frac{S}{K}$ for all $k \leq K$. So, for any fixed K, the variance is minimized by a balanced allocation $N_k = N$. Next, let $K^*(N) = \frac{B}{F+VN}$. So the variance simplifies to

$$W(K^*(N), N, \omega) = \frac{(F + VN)\rho}{B}\left(1 + \frac{\lambda}{N}\right)$$

First order condition returns

$$\frac{V\rho}{B}\left(1+\frac{\lambda}{N}\right)=\frac{(F+VN)\rho}{B}\left(\frac{\lambda}{N^2}\right)$$
$$V(N+\lambda)=\frac{(F+VN)\lambda}{N}$$
$$N^*(\omega)=N^*(\lambda)=\sqrt{\lambda\frac{F}{V}}$$

We further observe that $W(K^*(N),N,\omega)$ is strictly convex in N for $N\geq 0$, i.e.

$$W_{NN}(K^*(N),N,\omega)=-\frac{V\rho}{B}\frac{\lambda}{N^2}+\frac{V\rho\lambda}{BN^2}+\frac{(F+VN)\rho}{B}\frac{\lambda}{N^3}=\frac{(F+VN)\rho}{B}\frac{\lambda}{N^3}>0$$

Thus the local point satisfying the FOC is indeed the global minimum of the function. Finally, because $N^*(\lambda)$ is strictly monotonic in λ with $\lim_{\lambda\to 0}N^*(\lambda)=0$ and $\lim_{\lambda\to\infty}N^*(\lambda)=\infty$, the domain constraints on N lead to the final form of $N^*(\lambda)$. It follows that the optimal map (and oracle allocation) of the generalized problem is the same than the optimal map (and oracle allocation) of the simplified problem.

Oracle Variance. Assuming an interior solution for $N^*(\omega)$ (i.e. see Assumption 3.2),

$$K^*(\omega)=\frac{B}{F+\sqrt{FV\lambda}}$$

And,

$$W(K^*(\lambda),N^*(\lambda),\omega)=\frac{\rho}{K^*(\lambda)}\left(1+\frac{\lambda}{N^*(\lambda)}\right)=\frac{(F+\sqrt{FV\lambda})\rho}{B}\left(1+\sqrt{\lambda\frac{V}{F}}\right)$$
$$=\frac{\rho}{B}\left(F+2\sqrt{F\lambda V}+\lambda V\right)=\frac{\rho\left(\sqrt{F}+\sqrt{\lambda V}\right)^2}{B}$$

B Proofs of Section 4

Proof of Lemma 4.1. Unbiasedness of the estimators. To be used for a LLN argument,

$$\mathbb{E}[\tilde{w}(K,N)]=\frac{1}{K(N-1)}\sum_{k=1}^{K}\sum_{i=1}^{N}\mathbb{E}\left[(y_{ki}-\hat{\mu}^k(N))^2\right]=\frac{1}{K(N-1)}\sum_{k=1}^{K}\sum_{i=1}^{N}\mathrm{Var}(y_{ki}-\hat{\mu}^k(N))$$
$$=\frac{1}{K(N-1)}K(N-1)\sigma=\sigma$$
$$\mathbb{E}[\tilde{b}(K,N)]=\frac{1}{K-1}\sum_{k=1}^{K}\mathbb{E}\left[(\hat{\mu}^k(N)-\hat{\mu}(K,N))^2\right]=\frac{1}{K-1}(K-1)\left(\rho+\frac{\sigma}{N}\right)=\rho+\frac{\sigma}{N}$$
$$\mathbb{E}[\tilde{\rho}(K,N)]=\mathbb{E}[\hat{b}(K,N)]-\mathbb{E}\left[\frac{\hat{w}(K,N)}{N}\right]=\rho+\frac{\sigma}{N}-\frac{\sigma}{N}=\rho$$

Consistency,

$$\tilde{w}(K,N) = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{N-1}\sum_{i=1}^{N}(y_{ki}-\hat{\mu}^k(N))^2 \xrightarrow[K]{p} \sigma$$

where convergence follows from the LLN. We used cross-cluster independence to argue that that $\sum_{i=1}^{N}(y_{ki}-\hat{\mu}^k(N))^2 \perp\!\!\!\perp \sum_{i=1}^{N}(y_{li}-\hat{\mu}^l(N))^2$ for $k \neq l$. A similar argument holds for consistency of $\tilde{w}(K,N)$ under N asymptotics, namely

$$\begin{aligned}\tilde{w}(K,N) &= \frac{1}{N-1}\sum_{i=1}^{N}\frac{1}{K}\sum_{k=1}^{K}(y_{ki}-\hat{\mu}^k(N))^2 = \frac{1}{N-1}\sum_{i=1}^{N}\frac{1}{K}\sum_{k=1}^{K}(\varepsilon_{ki}-\bar{\varepsilon}_{ki})^2 \\ &= \frac{N}{N-1}\left[\frac{1}{N}\sum_{i=1}^{N}\varepsilon_{ki}^2-\bar{\varepsilon}_{ki}^2\right] \xrightarrow[N]{p} \sigma\end{aligned}$$

provided $\frac{1}{N}\sum_{i=1}^{N}\varepsilon_{ki}^2 \xrightarrow[N]{p} \sigma$ under the LLN given ε_{ki} is iid and $\bar{\varepsilon}_{ki}^2 \xrightarrow[N]{p} 0$ also under the LLN. Applying CMT completes the proof. Similarly,

$$\tilde{b}(K,N) = \frac{1}{K-1}\sum_{k=1}^{K}(\hat{\mu}^k(N)-\hat{\mu}(K,N))^2 \xrightarrow[K]{p} \rho + \frac{\sigma}{N}$$

where we used the same decomposition and argue that $\bar{u}_k + \bar{\bar{\varepsilon}}_{ki} \xrightarrow[K]{p} 0$. Consistency of $\tilde{\rho}(K,N)$ holds by the CMT and the consistency of $\tilde{b}(K,N)$ and $\tilde{w}(K,N)$ (only under K asymptotics). In addition, note that projection $\Pi_{[a,b]}(x)$ onto a closed interval $[a,b]$ is 1-Lipschitz, therefore convergence of $\mid \tilde{w}(K,N)-\sigma \mid$ implies convergence of $\mid \hat{w}(K,N)-\sigma \mid$. The same logic applies for $\hat{\rho}(K,N)$. Finally, consistency of $\hat{\lambda}(K,N)$ follows from consistency of $\hat{w}(K,N)$ and $\hat{\rho}(K,N)$ under CMT. □

B.1 `EtC-CS`

The proof of Theorem 4.3 follows from a set of preliminary Lemmas.

Lemma B.1.

Lemma B.1. *Performance of `EtC-CS`. For $K \geq 2, N \geq 2$, let $\hat{\lambda}(K,N)$ as defined in Equation 4.1 where $\hat{\lambda}_p = \hat{\lambda}(K^*(\lambda_M), N^*(\lambda_m))$. Then, under Assumption 3.2, and up to integer and terminal budget considerations, `EtC-CS` implements the design*

$$(K_{\texttt{EtC-CS}}, N_{\texttt{EtC-CS}}) = (K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$$

Therefore,

$$\Delta_{\texttt{EtC-CS}} = \frac{1}{2}W_{\lambda\lambda}(\omega_0)(\hat{\lambda}_p-\lambda_0)^2 + R_p$$

where $W_{\lambda\lambda}(\omega_0) \sim O(1/B)$ and $R_p \sim O\left((\hat{\lambda}_p-\lambda_0)^3\right)$.

Interpretation. When the pilot is selected small enough, the pilot prescribed optimal map is budget implementable. As a result, the excess variance is only local to λ, hence of second order.

Proof of B.1. The first part of the Lemma is a simple restatement of the algorithm under Assumption 3.2. The second part of the Lemma follows from a simple second order approximation around λ_0

$$
\begin{aligned}
\Delta_{\texttt{EtC-CS}} &= W(K_{\texttt{EtC-CS}}, N_{\texttt{EtC-CS}}, \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) \\
&= W(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p), \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) \\
&= W_\lambda(\omega_0)(\hat{\lambda}_p - \lambda_0) + \frac{1}{2} W_{\lambda\lambda}(\omega_0)(\hat{\lambda}_p - \lambda_0)^2 + O((\hat{\lambda}_p - \lambda_0)^3) \\
&= \frac{1}{2} W_{\lambda\lambda}(\omega_0)(\hat{\lambda}_p - \lambda_0)^2 + O((\hat{\lambda}_p - \lambda_0)^3)
\end{aligned}
$$

where the second equality follows from the first part of the Lemma and the last equality follows from optimality of (K^*, N^*) at λ_0 and the Envelope Theorem. Finally, we show analytically that $W_{\lambda\lambda}(\omega_0) \sim O(1/B)$

$$
\begin{aligned}
W(K^*(\lambda), N^*(\lambda), \omega_0) &= \rho_0 \frac{F + V\sqrt{\lambda \frac{F}{V}}}{B} \cdot \left(1 + \frac{\lambda_0}{\sqrt{\lambda \frac{F}{V}}}\right) = \frac{\rho_0}{B} \cdot \left(F + \lambda_0 V + \sqrt{VF}\left(\lambda_0/\lambda^{1/2} + \lambda^{1/2}\right)\right) \\
W_\lambda(K^*(\lambda), N^*(\lambda), \omega_0) &= \frac{\rho_0}{2B}\sqrt{FV}\left(-\lambda_0 \lambda^{-3/2} + \lambda^{-1/2}\right) \\
W_{\lambda\lambda}(K^*(\lambda), N^*(\lambda), \omega_0) &= \frac{\rho_0}{4B}\sqrt{FV}\left(3\lambda_0 \lambda^{-5/2} - \lambda^{-3/2}\right) \\
W_{\lambda\lambda}(K^*(\lambda_0), N^*(\lambda_0), \omega_0) &= \frac{\rho_0}{4B}\sqrt{FV}\left(3\lambda_0^{-3/2} - \lambda_0^{-3/2}\right) = \frac{\rho_0\sqrt{FV}}{2B\lambda_0^{3/2}}
\end{aligned}
$$

■

Lemma B.2.

Lemma B.2. *Upper Bound on $D_p^\lambda(\alpha)$. Let $K_p, N_p \geq 2$. Under Assumption 4.2, for any fixed $\alpha \in (0,1)$*

$$
\mathbb{P}(\mid \hat{\lambda}_p - \lambda \mid \leq D_p^\lambda(\alpha)) \geq 1 - \alpha
$$

where $D_p^\lambda(\alpha) \sim O(K_p^{-1/2})$.

Proof of Lemma B.2. The proof proceeds constructively. We proceed iteratively by constructing subexponential bounds for each of the elements entering the definition of $\hat{\lambda}$, i.e. $\tilde{w}(K_p, N_p), \tilde{b}(K_p, N_p)$ and $\tilde{\rho}(K_p, N_p)$ using the statistical properties of the errors. We then argue that the clipped versions of the estimators $\hat{w}(K, N)$ and $\hat{\rho}(K, N)$ satisfy the same subexponential bounds. Finally, we use these intermediate bounds to guarantee coverage over $\mid \hat{\lambda}(K_p, N_p) - \lambda \mid$. For the remainder of this proof, let $\tilde{a}_p = \tilde{a}(K_p, N_p)$ and $\hat{a}_p = \hat{a}(K_p, N_p)$.

Bounding $\mid \hat{w}_p - \sigma \mid$. We start by posing a few definitions. Let

$$
\varepsilon_k = (\varepsilon_{k1}, \ldots, \varepsilon_{kN_p})' \qquad M_{N_p} = I_{N_p} - \frac{1}{N_p}\mathbf{11'} \qquad \tilde{w}_p = \frac{1}{K_p}\sum_{k=1}^{K_p} \frac{1}{N_p - 1}\sum_{i}^{N_p}(\varepsilon_{li} - \bar{\varepsilon}_l)^2 = \frac{1}{K_p}\sum_{k=1}^{K_p}\frac{1}{N_p - 1}\varepsilon_k' M_{N_p} \varepsilon_k
$$

By Assumption 4.2, ε_k is a vector with independent subgaussian coordinates. Applying

Hanson-Wright inequality (see Rudelson and Vershynin (2013)), we obtain

$$\mathbb{P}(\mid \varepsilon_k' M_{N_p} \varepsilon_k - \mathbb{E}[\varepsilon_k' M_{N_p} \varepsilon_k] \mid \geq \alpha) \leq 2 \exp\left(-C \min\left\{\frac{\alpha^2}{\kappa_\varepsilon^2 \mid\mid M_{N_p} \mid\mid_F^2}, \frac{\alpha}{\kappa_\varepsilon \mid\mid M_{N_p} \mid\mid}\right\}\right)$$

where $|| \cdot ||_F$ is the Frobenius norm and $|| \cdot ||$ is the operator norm. Because M_{N_p} is an orthogonal projection of rank $N_p - 1$, $|| M_{N_p} ||_F^2 = (N_p - 1)$ and $|| M_{N_p} || = 1$. Moreover,

$$\mathbb{E}[\varepsilon_k' M_{N_p} \varepsilon_k] = \operatorname{tr}(M_{N_p} \mathbb{E}[\varepsilon_k \varepsilon_k']) = \sigma \operatorname{tr}(M_{N_p}) = (N_p - 1)\ \sigma$$

Therefore,

$$\begin{aligned}\mathbb{P}\left(\left|\frac{\varepsilon_k' M_{N_p} \varepsilon_k}{N_p - 1} - \frac{\mathbb{E}[\varepsilon_k' M_{N_p} \varepsilon_k]}{N_p - 1}\right| \geq \alpha\right) &= \mathbb{P}\left(\left|\frac{\varepsilon_k' M_{N_p} \varepsilon_k}{N_p - 1} - \sigma\right| \geq \alpha\right) \\ &\leq 2\exp\left(-C \min\left\{\frac{(N_p - 1)\alpha^2}{\kappa_\varepsilon^2}, \frac{(N_p - 1)\alpha}{\kappa_\varepsilon}\right\}\right)\end{aligned}$$

Finally, since ε_k vectors are independent across clusters

$$\mathbb{P}\left(|\tilde{w}_p - \sigma| \geq \alpha\right) \leq 2\exp\left(-C K_p \min\left\{\frac{(N_p - 1)\alpha^2}{\kappa_\varepsilon^2}, \frac{(N_p - 1)\alpha}{\kappa_\varepsilon}\right\}\right) \implies \mathbb{P}\left(\mid \tilde{w}_p - \sigma \mid \leq D_p^w(\alpha)\right) \geq 1 - \alpha/2$$

where $D_p^w(\alpha) = C\kappa_\varepsilon \left(\sqrt{\frac{\ln(4/\alpha)}{K_p(N_p - 1)}} + \frac{\ln(4/\alpha)}{K_p(N_p - 1)}\right)$ for some finite C. Under a 1-Lipschitz projection argument (as the one used in previous results), $\mid \hat{w}_p - \sigma \mid \leq \mid \tilde{w}_p - \sigma \mid$, therefore the same bound holds deterministically for $\mid \hat{w}_p - \sigma \mid$.

Bounding $\mid \tilde{b}_p - (\rho + \sigma/N_p) \mid$. A similar analysis holds for $\tilde{b}_p$. Let

$$v = (u_1 + \bar{\varepsilon}_1, \ldots, u_{K_p} + \bar{\varepsilon}_{K_p})' \qquad M_{K_p} = I_{K_p} - \frac{1}{K_p}\mathbf{1}\mathbf{1}' \qquad \tilde{b}_p = \frac{1}{K_p - 1} v' M_{K_p} v$$

By assumption v is a vector with independent $\sqrt{\kappa_u + \kappa_\varepsilon / N_p}$ subgaussian coordinates. Applying Hanson-Wright inequality, we obtain

$$\mathbb{P}(\mid v' M_{K_p} v - \mathbb{E}[v' M_{K_p} v] \mid \geq \alpha) \leq 2\exp\left(-C \min\left\{\frac{\alpha^2}{(\kappa_u + \kappa_\varepsilon/N_p)^2 \mid\mid M_{K_p} \mid\mid_F^2}, \frac{\alpha}{(\kappa_u + \kappa_\varepsilon/N_p) \mid\mid M_{K_p} \mid\mid}\right\}\right)$$

Under the same logic, $|| M_{K_p} ||_F^2 = (K_p - 1)$ and $|| M_{K_p} || = 1$. Moreover,

$$\mathbb{E}[v' M_{K_p} v] = (K_p - 1)\left(\rho + \frac{\sigma}{N_p}\right)$$

Therefore,

$$\mathbb{P}\left(\left|\frac{v' M_{K_p} v}{K_p - 1} - \frac{\mathbb{E}[v' M_{K_p} v]}{K_p - 1}\right| \geq \alpha\right) = \mathbb{P}\left(\left|\hat{b}_p - \left(\rho + \frac{\sigma}{N_p}\right)\right| \geq \alpha\right)$$
$$\leq 2\exp\left(-C \min\left\{\frac{(K_p - 1)\alpha^2}{(\kappa_u + \kappa_\varepsilon/N_p)^2}, \frac{(K_p - 1)\alpha}{\kappa_u + \kappa_\varepsilon/N_p}\right\}\right)$$

Finally,

$$\mathbb{P}\left(\mid \tilde{b}_p - (\rho + \sigma/N_p) \mid \leq D_p^b(\alpha)\right) \geq 1 - \alpha/2$$

where $D_p^b(\alpha) = C(\kappa_u + \frac{\kappa_\varepsilon}{N_p})\left(\sqrt{\frac{\ln(4/\alpha)}{K_p - 1}} + \frac{\ln(4/\alpha)}{K_p - 1}\right)$.

Bounding $\mid \hat{\rho}_p - \rho \mid$. Applying the triangle inequality,

$$\mid \tilde{\rho}_p - \rho \mid = \mid \tilde{b}_p - \tilde{w}_p/N_p - \rho \mid = \mid \tilde{b}_p - (\rho + \sigma/N_p) - 1/N_p(\tilde{w}_p - \sigma) \mid \leq \mid \tilde{b}_p - (\rho + \sigma/N_p) \mid + \frac{1}{N_p} \mid \tilde{w}_p - \sigma \mid$$

$\{\mid \tilde{b}_p - (\rho + \sigma/N_p) \mid \leq D_p^b(\alpha)\}$ and $\{\mid \tilde{w}_p - \sigma \mid \leq D_p^w(\alpha)\}$ each hold with probability $1 - \alpha/2$, therefore $\mathbb{P}(\mid \tilde{\rho}_p - \rho \mid \leq D_p^\rho(\alpha)) \geq 1 - \alpha$, where $D_p^\rho(\alpha) = D_p^b(\alpha) + \frac{1}{N_p} D_p^w(\alpha)$. Under the same 1-Lipschitz projection argument, the same bound holds deterministically for $\mid \hat{\rho}_p - \rho \mid$.

Bounding $\mid \hat{\lambda}_p - \lambda \mid$. Observe

$$\mid \hat{\lambda}_p - \lambda \mid = \left|\frac{\hat{w}_p}{\hat{\rho}_p} - \frac{\sigma}{\rho}\right| = \left|\frac{\rho\ \hat{w}_p - \sigma\ \hat{\rho}_p}{\rho\ \hat{\rho}_p}\right| = \left|\frac{\rho\ (\hat{w}_p - \sigma) - \sigma\ (\hat{\rho}_p - \rho)}{\rho\ \hat{\rho}_p}\right| \leq \left|\frac{\rho\ (\hat{w}_p - \sigma) - \sigma\ (\hat{\rho}_p - \rho)}{\rho\ \rho_m}\right|$$

With probability $1 - \alpha$, the bounds for $\hat{w}_p$ and $\hat{\rho}_p$ hold simultaneously, hence

$$\mathbb{P}\left(\mid \hat{\lambda}_p - \lambda \mid \leq D_p^\lambda(\alpha)\right) \geq 1 - \alpha$$

where $D_p^\lambda(\alpha) = \frac{D_p^w(\alpha)}{\rho_m} + \frac{\sigma_M D_p^\rho(\alpha)}{\rho_m^2}$. The final rate then matches the rate of $\hat{\rho}_p$, i.e.

$$D_p^\lambda(\alpha) \sim O\left(K_p^{-1/2}\right)$$

Observe that D_p^w decreases much faster, at approximately $O\left(\frac{1}{K_p(N_p - 1)}\right)$. ■

Proof of Theorem 4.3. Define the good event $\mathcal{E} := \{\mid \hat{\lambda}_p - \lambda_0 \mid \leq D_p^\lambda(\alpha)\}$. Under the event $\mathcal{E}$,

$$
\begin{aligned}
\Delta_{\texttt{EtC-ACS}} &= \frac{C_1}{B}(\hat{\lambda}_p - \lambda_0)^2 + R_p \leq \frac{C_1}{B} D_p(\alpha)^2 + R_p \\
&\leq \frac{C_2}{B}\frac{\ln(1/\alpha)}{K_p} + R_p \leq \frac{C_2}{B}\frac{\ln(1/\alpha)}{K^*(\lambda_M)} + R_p \\
&\leq C_3 \frac{\ln(1/\alpha)}{B^2}
\end{aligned}
$$

where the first equality follows from Lemma B.1, the second and third inequalities follows from Lemma B.2 under the good event $\mathcal{E}$, the fourth equality by definition of $K_p = K^*(\lambda_M)$ under `EtC-CS`, and the fifth inequality by $K^*(\lambda_M) = \frac{B}{F+VN^*(\lambda_M)}$. Moreover, we note that $R_p \sim O(B^{-5/2})$, thus this term gets absorbed by the constant C_3. We note that both C_2 and C_3 depend on N_p, but not on B, provided $N_p = \sqrt{\lambda_m F/V} \perp\!\!\!\perp B$. Theorem follows from noting that event $\mathcal{E}$ has probability greater or equal than $1-\alpha$ as per Lemma B.2. □

B.2 `ACS`

Similar to the construction of the proof of Theorem 4.3, the proof of Theorem 4.4 follows from a set of preliminary Lemmas.

Lemma B.3.

Lemma B.3. *Performance of* `ACS`*. Let τ be the stopping time induced by Algorithm 2, and define $\hat{\lambda}_\tau = \hat{\lambda}(\tau, N_p)$ where $N_p = N^*(\lambda_m)$. Then, under Assumption 3.2, and up to integer and terminal budget considerations,* `ACS` *implements the design*

$$
(K_{ACS}, N_{ACS}) = (K^*(\hat{\lambda}_{\tau-1}), N^*(\hat{\lambda}_{\tau-1}))
$$

Therefore,

$$
\Delta_{ACS} = \frac{1}{2} W_{\lambda\lambda}(\omega_0)(\hat{\lambda}_{\tau-1} - \lambda_0)^2 + R_{\tau-1}
$$

where $W_{\lambda\lambda}(\omega_0) \sim O(1/B)$ and $O\left((\hat{\lambda}_{\tau-1} - \lambda_0)^3\right)$.

Interpretation. The logic is exactly parallel to the one in Lemma B.1. When, N_p is selected small enough, `ACS` stops (i) before exhausting the budget and (ii) before oversampling clusters (according to $\hat{\lambda}_{\tau-1}$. As a result, the algorithm can successfully implement the optimal map as prescribed by $\hat{\lambda}_{\tau-1}$, and the excess variance is only local to λ, hence of second order.

Proof of B.3. The first part of the Lemma is immediate. By the estimator's clipping $N_p = N^*(\lambda_m) \leq N^*(\hat{\lambda}_K)$ for all K (including at time $\tau - 1$ and τ). It follows that for some stopping time τ, $K^*(\hat{\lambda}_\tau) = \frac{B}{F+VN^*(\hat{\lambda}_\tau)} \leq \frac{B}{F+VN_p}$. Therefore, up to integer considerations, $\tau = \inf\{K : K+1 > K^*(\hat{\lambda}_K)\}$. In other words, the algorithm stops before exhausting the budget. Moreover, at time $\tau - 1$, $\tau \leq K^*(\hat{\lambda}_{\tau-1})$, otherwise the algorithm would have not explored cluster τ. It follows that the design $(K^*(\lambda_{\tau-1}), N^*(\hat{\lambda}_{\tau-1}))$ is budget feasible (provided $\tau \leq K^*(\hat{\lambda}_{\tau-1})$ and $N_p \leq N^*(\hat{\lambda}_{\tau-1})$). The second part of the Lemma follows the same second order approximation argument as in Lemma B.1). ■

Lemma B.4.

Lemma B.4. *Upper Bound on $D_k^{\lambda}(\alpha)$. Let $K_m = K^*(\lambda_m), K_M = K^*(\lambda_M)$ and fix $N_p \geq 2$. Under Assumption 3.2 and Assumption 4.2, for any fixed $\alpha \in (0,1)$*

$$\mathbb{P}(\mid \hat{\lambda}_K - \lambda \mid \leq D_K^{\lambda}(\alpha) \ \forall \ K \in [K_M, K_m]) \geq 1 - \alpha$$

where $D_K^{\lambda}(\alpha) \sim O\left(\frac{\ln(K_m)}{K}\right)$.

Proof of Lemma B.4. The proof parallels the construction in Lemma B.2 with the only difference that bounds over the intermediate elements $\tilde{w}(K, N_p), \tilde{b}(K, N_p)$ and $\tilde{\rho}(K, N_p)$ must hold uniformly over $K \in [K_M, K_m]$, which is compact under Assumption 3.2. We avoid repetition in the argument whenever possible and encourage the reader to consider this proof together with the derivations in Lemma B.2. For the remainder of this proof, let $\tilde{a}_K = \tilde{a}(K, N_p)$ and $\hat{a}_K = \hat{a}(K, N_p)$.

Starting from

$$\mathbb{P}\left(|\tilde{w}_K - \sigma| \geq \alpha\right) \leq 2\exp\left(-CK\min\left\{\frac{(N_p-1)\alpha^2}{\kappa_\varepsilon^2}, \frac{(N_p-1)\alpha}{\kappa_\varepsilon}\right\}\right)$$

apply the union bound over $K \in [K_M, K_m]$ to obtain

$$\mathbb{P}\left(\mid \tilde{w}_K - \sigma \mid \leq D_K^w(\alpha) \ \ \forall \ K_M \leq K \leq K_m\right) \geq 1 - \alpha/2$$

with $D_K^w(\alpha) = C\kappa_\varepsilon\left(\sqrt{\frac{\ln(4K_m/\alpha)}{K(N_p-1)}} + \frac{\ln(4K_m/\alpha)}{K(N_p-1)}\right)$ for some finite C. Under a 1-Lipschitz projection argument $\mid \hat{w}_K - \sigma \mid \leq \mid \tilde{w}_K - \sigma \mid$ for all K, therefore the same bound holds deterministically for the event $\mid \tilde{w}_K - \sigma \mid \leq D_K^w(\alpha) \ \ \forall \ K_M \leq K \leq K_m$.

The same logic applies to $\tilde{b}_K$ and $\tilde{\rho}_K$, what allows us to recover a uniform bound of the form

$$\mathbb{P}\left(\mid \hat{b}_K - (\rho + \sigma/N_p) \mid \leq D_K^b(\alpha) \ \ \forall \ K \in [K_M, K_m]\right) \geq 1 - \alpha/2$$

$$\mathbb{P}(\mid \hat{\rho}_K - \rho \mid \leq D_K^{\rho}(\alpha) \ \forall \ K \in [K_M, K_m]) \geq 1 - \alpha$$

with $D_K^b(\alpha) = C(\kappa_u + \frac{\kappa_\varepsilon}{N_p})\left(\sqrt{\frac{\ln(4K_m/\alpha)}{K-1}} + \frac{\ln(4K_m/\alpha)}{K-1}\right)$, and $D_K^{\rho}(\alpha) = D_K^b(\alpha) + \frac{1}{N_p}D_K^w(\alpha)$. Finally,

$$\mathbb{P}\left(\mid \hat{\lambda}_K - \lambda \mid \leq D_K^{\lambda}(\alpha)\right) \geq 1 - \alpha$$

where $D_K^{\lambda}(\alpha) = \frac{D_K^w(\alpha)}{\rho_m} + \frac{\sigma_M D_K^{\rho}(\alpha)}{\rho_m^2}$, with the final rate matching that of $\hat{\rho}_K$, i.e.

$$D_K^{\lambda}(\alpha) \sim O\left(\sqrt{\frac{\ln K_m}{K}}\right)$$

Observe that D_K^w decreases much faster, at approximately $O\left(\frac{\ln K_m}{K(N_p-1)}\right)$. $\square$

Lemma B.5.

Lemma B.5. *Linear Stopping Time. Let τ be the stopping time induced by* `ACS` *according to Equation 4. Then, under Assumption 3.2, τ is $\Theta(B)$ with probability 1.*

Proof of Lemma B.5. The proof is almost immediate. Note that $K^*(\lambda) = B\kappa(\lambda)$ with $\kappa(\lambda)$ strictly decreasing in λ. Then,

- Recall $K_m = B\kappa(\lambda_m)$ and $K_m \geq K^*(\hat{\lambda}_K)$ for all $K \in [K_M, K_m]$. In particular, $K_m \geq K^*(\hat{\lambda}_{K_m-1})$, so the algorithm should halt at $K_m - 1$ the latest. Therefore, $K_m \geq \tau$.
- Recall $K_M = B\kappa(\lambda_M)$ and $K_M - 1 \leq K^*(\hat{\lambda}_K) - 1$ for all $K \in [K_M, K_m]$. Moreover, $K + 1 \leq K^*(\hat{\lambda}_K)$ for all $K \leq K_M - 1$, so the algorithm will not halt at any $K \leq K_M - 1$. Therefore, $K_M - 1 \leq \tau$.

Finally, under Assumption 3.2, $K_M - 1 \geq 1$, so the interval is separated from 0, i.e. $1 \leq K_M - 1 \leq \tau \leq K_m$. Proof is concluded by recalling that both K_M and K_m are $\Theta(B)$. $\blacksquare$

Proof of Theorem 4.4. Define the good event $\mathcal{E} := \{\mid \hat{\lambda}_K - \lambda_0 \mid \leq D_K^\lambda(\alpha)\ \forall K \in [K_M, K_m]\}$. Under the event $\mathcal{E}$,

$$\begin{aligned}
\Delta_{\texttt{ACS}} &= \frac{C_1}{B}(\hat{\lambda}_{\tau-1} - \lambda_0)^2 + R_{\tau-1} \leq \frac{C_1}{B} D_{\tau-1}(\alpha)^2 + R_{\tau-1} \\
&\leq \frac{C_2}{B}\frac{\ln(K_m/\alpha)}{\tau - 1} + R_{\tau-1} \leq \frac{C_3}{B}\frac{\ln(K_m/\alpha)}{B} + R_{\tau-1} \\
&\leq C_4 \frac{\ln(B/\alpha)}{B^2}
\end{aligned}$$

where the first equality follows from Lemma B.3, the second and third inequality follow from Lemma B.4 under the good event $\mathcal{E}$, the fourth inequality from Lemma B.5, and the fifth inequality by definition of K_m. Moreover, we note that $R_{\tau-1} \sim O((\ln B)^{3/2}/B^{5/2})$, thus this term get absorbed by the constant C_4. Theorem follows from noting that event $\mathcal{E}$ has probability greater or equal than $1 - \alpha$ as per Lemma B.4. $\square$

B.3 Worst Case and Expected Excess Variance

Proof of Lemma 4.5. For all policies satisfying the budget constraint, the variance minimization problem admits a representation as a single variable unconstrained minimization exercise of the form $\min_{2 \leq N \leq (B-2F)/(2V)} W(K^*(N), N, \omega)$. As discussed in Appendix A, $W(K^*(N), N, \omega)$ is convex in N, therefore the maximum is achieved at the boundaries, i.e. $W(K^*(2), 2, \omega)$ or $W(2, (B-2F)/(2V), \omega)$. Our results would not change materially if the designer could select $N = 1$ or $N = (B - F)/V$ (i.e. $K = 1$). However, for consistency with the rest of the paper we keep the same domain. Now, simple calculations show that

$$W(K^*(2),2,\omega_0) - W(K^*(\lambda_0),N^*(\lambda_0),\omega_0) = \frac{\rho_0}{2B}\left(\sqrt{\lambda_0 F} - 2\sqrt{V}\right)^2$$
$$W(2,(B-2F)/(2V),\omega_0) - W(K^*(\lambda_0),N^*(\lambda_0),\omega_0) = \frac{\rho_0}{2}\left(1 + \frac{2\lambda_0 V}{B-2F}\right) - \frac{\rho_0}{B}\left(\sqrt{F} + \sqrt{\lambda_0 V}\right)^2$$

While the first term converges to 0 as $B \to \infty$, the second term converges to $\frac{\rho_0}{2}$ from above. Therefore, the worst case excess variance is asymptotically upper bounded by $\frac{\rho_0}{2} + \delta$ for some small positive δ. □

Proof of Theorem 4.6. Set $\alpha = \frac{C_\alpha}{B^2}$ for some finite C_α

$$\begin{aligned}
\mathbb{E}_{\texttt{EtC-CS}}\left[\Delta_{\texttt{EtC-CS}}\right] &= \mathbb{P}_{\texttt{EtC-CS}}(\mathcal{E}) \cdot \mathbb{E}_{\texttt{EtC-CS}}\left[\Delta_{\texttt{EtC-CS}} \mid \mathcal{E}\right] + \mathbb{P}_{\texttt{EtC-CS}}(\overline{\mathcal{E}}) \cdot \mathbb{E}_{\texttt{EtC-CS}}\left[\Delta_{\texttt{EtC-CS}} \mid \overline{\mathcal{E}}\right] \\
&\leq \alpha\left(\frac{\rho_M}{2} + \delta\right) + C\frac{\ln(1/\alpha)}{B^2} \\
&= \frac{C_\alpha}{B^2} + C\frac{\ln(B^2/C_\alpha)}{B^2} \leq C_1\frac{\ln(B)}{B^2}
\end{aligned}$$

where the key inequality follows from Theorem 4.3 and Lemma 4.5. α has been selected such that the second expression is minimized. In principle, C depends only on known parameters of the model (including $F, V, \Omega, \kappa_\varepsilon, \kappa_u$), so the designer can set $\alpha = C/B^2$ for explicit performance guarantees. Similarly,

$$\begin{aligned}
\mathbb{E}_{\texttt{ACS}}\left[\Delta_{\texttt{ACS}}\right] &= \mathbb{P}_{\texttt{ACS}}(\mathcal{E}) \cdot \mathbb{E}_{\texttt{ACS}}\left[\Delta_{\texttt{ACS}} \mid \mathcal{E}\right] + \mathbb{P}_{\texttt{ACS}}(\overline{\mathcal{E}}) \cdot \mathbb{E}_{\texttt{ACS}}\left[\Delta_{\texttt{ACS}} \mid \overline{\mathcal{E}}\right] \\
&\leq \alpha\left(\frac{\rho_M}{2} + \delta\right) + C\frac{\ln(B/\alpha)}{B^2} \\
&= \frac{C_\alpha}{B^2} + C\frac{\ln(B^3/C_\alpha)}{B^2} \leq C_2\frac{\ln(B)}{B^2}
\end{aligned}$$

where the key inequality follows from Theorem 4.4 and Lemma 4.5. □

C Proofs of Section 5

C.1 Lower Bound Derivation

Policy Class. We first define the relevant policy class Π. The policy class is fairly general and contains most reasonable policies as especial cases. In particular, we consider policies which, in every period t, select a cluster $k_t = k$ and sample at least one unit $N_{k_t,t} \geq 1$ until they run out of budget. As a result, the game contains at most $T = K_m N_M$ rounds with strictly positive sampling. If no additional observation can be sampled without violating the budget in period τ, $N_{k_t,t} = 0$ for all $t \geq \tau$. Importantly, the same cluster k can be revisited several times. Let $N_k(T) = \sum_{t=1}^{T} \mathbb{1}(k_t = k) N_{k_t,t}$ be the number of samples from cluster k up to period T. Define as well $K(T) = \sum_k \mathbb{1}(N_k(T) > 0)$. All policies must satisfy the budget constraint almost surely, i.e.

$$\mathbb{P}_\pi \left(\sum_{k\in[K(T)]}^{K} (F + VN_k(T)) \leq B \right) = 1$$

Data Generating Process. Consider two normally distributed Data Generating Processes (DGPs) y_-, y_+ indexed by ω_- and ω_+, where $\omega_- = (\mu, \rho_-, \lambda_-)$ and $\omega_+ = (\mu, \rho_+, \lambda_+)$ where $\lambda_- = \lambda_0 - \delta$, $\lambda_+ = \lambda_0 + \delta$ for some $\lambda_0 \in (\lambda_m, \lambda_M)$ and a small positive δ. Additionally, $\rho_- = \sigma/\lambda_-$ and $\rho_+ = \sigma/\lambda_+$. Under the additive errors model in Equation 1

$$y_{+ki} = \mu + u_{+k} + \varepsilon_{ki}$$

where $u_{+k} \sim N(0, \rho_+)$. y_{-ki} and u_{-k} are defined as expected.

Separating Events. As it is common practice in lower bound derivations, we characterize a separating event. Following the characterization of the variance for the generalized game (where $N_k \neq N$ for all k), we write the separating event in terms of deviations from the harmonic mean N_H. To do so, we need to characterize (or at least lower bound) the excess variance in terms of the harmonic mean. By the linearity of the cost structure $K_T(F + \sum_{k\in[K_T]} VN_k(T)) = K_T(F + V\bar{N}(T))$ where $\bar{N} = \frac{1}{K_T}\sum_{k\in[K_T]} N_k(T)$ is the arithmetic mean. We know that $N_H(T) \leq \bar{N}(T)$, hence $\frac{1}{K_T} \geq \frac{F+VN_H}{B}$. It follows

$$W(\boldsymbol{N}(T), \omega) = \frac{\rho}{K_T(\boldsymbol{N}(T))}\left(1 + \frac{\lambda}{N_H(\boldsymbol{N}(T))}\right) \geq \frac{\rho(F + VN_H(T))}{B}\left(1 + \frac{\lambda}{N_H(T)}\right) = W_H(N_H(T), \omega)$$

Now, take the environment $\omega = \omega_+$, let $N_+ = N^*(\lambda_+)$ and consider the event $N_H \leq N_0 = N^*(\lambda_0)$. Under the remarks above, the excess variance of a policy which selects $\boldsymbol{N}(T)$ such that $N_H \leq N_0$ can be lower bounded by

$$W(\boldsymbol{N}(T), \omega_+) - W(N_+, \omega_+) \geq W_H(N_H, \omega_+) - W_H(N_H, \omega_+) \geq W_H(N_0, \omega_+) - W_H(N_+, \omega_+)$$

where we used the fact that $W(N^*, \omega) = W_H(N^*, \omega)$ because the oracle allocation is a balanced one (under a simple convexity argument). A symmetric result holds for ω_-. Finally, by noting that the expression for W_H is completely equivalent to the expression for $W(K^*(N), N, \omega)$ in Appendix A, one can recover a second order Taylor approximation of the form

$$W_H(N_0, \omega_+) - W_H(N_+, \omega_+) = \frac{\rho_0\sqrt{FV}}{4B\lambda_0^{3/2}}(\lambda_0 - \lambda_+)^2 + O((\lambda_0 - \lambda_+)^3/B)$$

Augmented Experiment and Divergence Decomposition. For some policy $\pi \in \Pi$, let $(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the environments y_- and y_+, respectively. Now, consider the augmented experiment where, every time the learner samples cluster k, she not only observes $(y_{ki})_{N_k}$ but also u_k. Let $(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the augmented environments. By data processing,

$$D_{\mathrm{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \leq D_{\mathrm{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$$

The augmented game simplifies the analysis of the KL-divergence significantly, as revisiting a cluster does not add any more *cluster-level* KL. This fact, together with the zero KL contribution of individual errors (provided they have the same distribution in both environments), allows us to decompose $D_{\mathrm{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$ in the usual *bandit style*,[22] i.e.

$$D_{\mathrm{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) = \mathbb{E}_{\pi-}\left[K_T\right] D_{\mathrm{KL}}(N(0, \rho_-), N(0, \rho_+))$$

We further observe that for all policies $\pi \in \Pi$, $FK_T + \sum_k VN_k(T) \leq B$ a.s., thus $K_T \leq B/F$ a.s.. The expression above incorporates the fact that all clusters within each environment are equally distributed. Next

$$\begin{aligned} D_{\mathrm{KL}}(N(0, \rho_-), N(0, \rho_+)) &= \frac{1}{2}\left[\frac{\rho_-}{\rho_+} - 1 - \ln\left(\frac{\rho_-}{\rho_+}\right)\right] = \frac{1}{2}\left[\frac{\lambda_+}{\lambda_-} - 1 - \ln\left(\frac{\lambda_+}{\lambda_-}\right)\right] \\ &= \frac{1}{2}\phi\left(\frac{\lambda_+}{\lambda_-}\right) = \frac{1}{2}\phi\left(1 + \frac{2\delta}{\lambda_0 - \delta}\right) \end{aligned}$$

where the third equality serves as definition of $\phi(x) = x - 1 - \ln(x)$. For $\delta \leq \lambda_0/2$, and using the fact that $\phi(1+x) \leq x^2/2$

$$\frac{1}{2}\phi\left(\frac{\lambda_+}{\lambda_-}\right) \leq \frac{1}{2}\phi\left(1 + \frac{4\delta}{\lambda_0}\right) \leq \frac{4\delta^2}{\lambda_0^2}$$

Therefore,

$$D_{\mathrm{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \leq \frac{B}{F}\frac{4\delta^2}{\lambda_0^2} = C_1 B \delta^2$$

The argument is now completed through a standard Bretagnolle-Huber argument

[22] See Chapter 15 in Lattimore and Szepesvári (2020) for a complete treatment.

$$
\begin{aligned}
&\tilde{\mathbb{E}}_{\pi-}\left[W(K,\boldsymbol{N},\omega_-)\right] - W(K^*(\lambda_-),N^*(\lambda_-),\omega_-) + \tilde{\mathbb{E}}_{\pi+}\left[W(K,\boldsymbol{N},\omega_+)\right] - W(K^*(\lambda_+),N^*(\lambda_+),\omega_+) \geq \\
&\tilde{\mathbb{E}}_{\pi-}\left[W_H(N_H,\omega_-)\right] - W_H(N_-,\omega_-) + \tilde{\mathbb{E}}_{\pi+}\left[W_H(N_H,\omega_+)\right] - W_H(N_+,\omega_+) \geq \\
&\tilde{\mathbb{P}}_{\pi-}\left(N_H > \sqrt{\lambda_0\frac{F}{V}}\right)\cdot\left(\mathbb{E}_{\pi-}\left[W_H(N_H,\omega_-)\mid N_H > \sqrt{\lambda_0\frac{F}{V}}\right] - W_H(N_-,\omega_-)\right) + \\
&\quad\tilde{\mathbb{P}}_{\pi+}\left(N_H \leq \sqrt{\lambda_0\frac{F}{V}}\right)\cdot\left(\mathbb{E}_{\pi+}\left[W(N_H,\omega_+)\mid N_H \leq \sqrt{\lambda_0\frac{F}{V}}\right] - W(N_+,\omega_+)\right) = \\
&\frac{\rho_0\sqrt{FV}}{4B\lambda_0^{3/2}}\delta^2\left(\tilde{\mathbb{P}}_{\pi-}\left(N_H > \sqrt{\lambda_0\frac{F}{V}}\right) + \tilde{\mathbb{P}}_{\pi+}\left(N_H \leq \sqrt{\lambda_0\frac{F}{V}}\right)\right) + O(\delta^3/B) \geq \\
&\frac{\rho_0\sqrt{FV}}{4B\lambda_0^{3/2}}\delta^2\frac{1}{2}\exp\left(-D_{\mathrm{KL}}(\tilde{\mathbb{P}}_{\pi-},\tilde{\mathbb{P}}_{\pi+})\right) - \frac{C_3\delta^3}{B} \geq \\
&\frac{\rho_0\sqrt{FV}}{4B\lambda_0^{3/2}}\delta^2\frac{1}{2}\exp\left(-D_{\mathrm{KL}}(\mathbb{P}_{\pi-},\mathbb{P}_{\pi+})\right) - \frac{C_3\delta^3}{B} \geq \\
&\frac{\rho_0\sqrt{FV}}{8B\lambda_0^{3/2}}\delta^2\exp\left(-C_1\delta^2 B\right) - \frac{C_2\delta^3}{B}
\end{aligned}
$$

Where the first inequality follows from our variance relaxation under the harmonic depth, the second inequality is a simple event partition, the third inequality follows from the second order Taylor expansion anticipated in the Section, the fourth inequality follows from Bretagnole-Huber inequality, the fifth inequality from a data processing argument, and the final inequality from our derivations above. Finally, by selecting $\delta = \sqrt{1/(C_1B)}$, which is smaller than $\lambda_0 \leq \lambda/2$ for sufficiently large B, we recover

$$
\frac{\rho_0\sqrt{FV}}{8B\lambda_0^{3/2}}\frac{1}{C_1B}\exp(-1) - \frac{C_2\delta^3}{B} \geq \frac{C_3}{B^2} - \frac{C_4}{B^{5/2}}
$$

for some finite and strictly positive C_3. Result follows from applying the inequality $\max(a,b) \geq \frac{a+b}{2}$ and absorbing it (as well as the higher order terms) into a constant C. $\square$

C.2 Policy Comparison

Proof of Lemma 5.2. The first part of the Lemma is immediate, and follows from a simple second order approximation around λ_0. See Appendix A for a complete derivation. For the minmax optimal policy, recall that

$$
\begin{aligned}
W(K^*(\lambda),N^*(\lambda),\omega_0) - W(K^*(\lambda_0),N^*(\lambda_0),\omega_0) &= \frac{\rho_0}{B}\left[F + V\lambda_0 + \sqrt{FV}\left(\sqrt{\lambda} + \frac{\lambda_0}{\sqrt{\lambda}}\right)\right] - \frac{\rho_0(\sqrt{F}+\sqrt{V\lambda_0})^2}{B} \\
&\frac{\rho_0\sqrt{FV}}{B}\cdot\frac{\left(\sqrt{\lambda}-\sqrt{\lambda_0}\right)^2}{\sqrt{\lambda}}
\end{aligned}
$$

Hence, the maximum is achieved at $\rho_0 = \rho_M$ and $\lambda_0 \in \{\lambda_m, \lambda_M\}$ (whichever yields a higher loss). Finally, the minimum is obtained by

$$\min_{\lambda} \max \left\{ \frac{\left(\sqrt{\lambda} - \sqrt{\lambda_M}\right)^2}{\sqrt{\lambda}}, \frac{\left(\sqrt{\lambda} - \sqrt{\lambda_m}\right)^2}{\sqrt{\lambda}} \right\}$$

which yields $\lambda_{mM} = \left(\frac{\sqrt{\lambda_m}+\sqrt{\lambda_M}}{2}\right)^2$. Finally, the same second order approximation returns

$$W(K^*(\lambda_{mM}), N^*(\lambda_{mM}), \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) = \frac{C(\lambda_{mM} - \lambda_0)^2}{B} + O\left(\frac{\mid \lambda_{mM} - \lambda_0 \mid^3}{B}\right) \sim O(1/B)$$

D Proofs of Section 6

Assumption D.1. *Inference Assumptions*

1. ***Symmetry and Independent Errors:*** *Let $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ be the vector of cluster k errors. Assume $\mathbb{E}[u_k] = \mathbb{E}[\varepsilon_{ki}] = 0$, $Var(u_k) = \rho$, $Var(\varepsilon_{ki}) = \sigma$, $C_k \overset{d}{=} -C_k$ with C_k iid and $u_k \perp\!\!\!\perp \varepsilon_{ki}$ for all i, k.*

2. ***Finite Fourth Moments:*** *For some small positive $\eta > 0$, $\mathbb{E}[\mid u_k \mid^{4+\eta}] < \infty$, $\mathbb{E}[\mid \varepsilon_{ki} \mid^{4+\eta}] < \infty$*

3. ***Consistent Exploration and Consistent Design:*** *Let (K_A, N_A) be the design induced by some algorithm π, where $K_A = K^*(\hat{\lambda}(K_E, N_p)), N_A = N^*(\hat{\lambda}(K_E, N_p))$ for some fixed N_p and some cluster exploration sample K_E determined by the same algorithm π. Recall $\kappa(\lambda) = K^*(\lambda)/B$ for some $\lambda \in [\lambda_m, \lambda_M]$. Let $\kappa_0 = \kappa(\lambda_0)$. Under B asymptotics, assume that $K_E/B \to \kappa_E > 0$ uniformly in π. Further assume that $\hat{\lambda}(K_E, N_p) - \lambda_0 = O_p(B^{-1/2})$*

Comment on Assumption D.1. The only distinct assumption in Assumption i), which imposes a symmetry assumption on the errors. This assumption is comparable to symmetry assumptions in randomization inference (see Canay et al. (2017)). Assumption ii) is strong but also standard in inference involving variances. Finally, Assumption iii) is satisfied for `EtC-CS` and `ACS`. In particular, the Assumption is satisfied in `EtC-CS` for deterministic $K_E = K_M$ and in `ACS` for $K_E = K^*(\hat{\lambda}_{\tau-1})$ which (i) is linear in B under Lemma B.5 and (ii) converges uniformly to $K^*(\lambda_0)$ under B asymptotics.[23] Finally, under Assumption iii), the design (K_A, N_A) is consistent for $K^*(\lambda_0), N^*(\lambda_0)$ (see Section 4 and Appendix B for further discussion).

Set-Up. Define

$$\bar{\varepsilon}_k^N = \frac{1}{N}\sum_{i=1}^{N} \varepsilon_{ki} \qquad W_k = \frac{1}{N_p - 1}\sum_{i=1}^{N_p} (\varepsilon_{ki} - \bar{\varepsilon}_k^{N_p})^2 \qquad \hat{w}_K = \frac{1}{K-1}\sum_{k=1}^{K} W_k$$

[23]Somewhat confusingly, $K_A = K_E$ under `ACS` (but not under `EtC-CS`) as no new cluster is explored after the pilot. Recall the discussion in Section 4

where $\mathbb{E}[W_k] = \sigma$ and $A_k = W_k - \sigma$. In particular, note that W_k is the usual within-cluster k variance and $\hat{w}_K$ is the average of within-cluster variances, which also identifies the within-cluster variance for every k (as in Equation 4.1). Now, observe that

$$\hat{\mu}(K,N) - \mu_0 = \frac{1}{K}\sum_{k=1}^{K} M_{k,N} \qquad M_{k,N} = u_k + \frac{1}{N}\sum_{i=1}^{N}\varepsilon_{ki} \qquad \bar{M}_{K,N} = \frac{1}{K}\sum_{k=1}^{K} M_{k,N}$$

where $M_{k,N}$ is the average of cluster k errors such that $\mathbb{E}[M_{k,N}] = 0$. Similarly, just like we did with the within-cluster variance $\hat{w}_k$,

$$\hat{b}_K = \frac{1}{K-1}\sum_{k=1}^{K}(M_{k,N_p} - \bar{M}_{K,N_p})^2$$

where $\hat{b}_K$ is the between-cluster variance (as in Equation 4.1) such that $\mathbb{E}[\hat{b}_K] = \rho + \sigma/N_p = \theta_{N_p}$. In addition, define the centered second moment of the average of cluster k errors

$$B_k = M^2_{k,N_p} - \theta_{N_p}$$

Unlike $\hat{w}_k$, $\hat{b}_k$ cannot be written as the average of B_k as $\hat{b}_k$ demeans M_{k,N_p} by the (residuals of the) grand mean $\bar{M}_{k,N_p}$.

Preliminary Lemmas

Lemma D.2. *Conditional Zero Mean. Define* $L_k = \frac{\theta_{N_p}}{\rho^2}A_k - \frac{\sigma}{\rho^2}B_k$

$$\mathbb{E}[M_{k,N} \mid L_k] = 0 \;\; a.s$$

Proof of Lemma D.2. Consider the cluster error sign map $f(\{C_k\}) \mapsto f(\{-C_k\})$ for all k. Then, $M_{k,N} \mapsto -M_{k,N}$ (so $M_{k,N}$ is odd), while L_k provided $W_k \mapsto W_k$ and $B_k \mapsto B_k$, so they are even. Note that B_k is even as it is the square of an odd function minus a constant. Then, for any bounded ϕ

$$\mathbb{E}[M_{k,N}\phi(L_k)] = \mathbb{E}[-(M_{k,N})\phi(L_k)] = -\mathbb{E}[M_{k,N}\phi(L_k)] \implies \mathbb{E}[M_{k,N}\phi(L_k)] = 0$$

where the first equality follows from Assumption D.1.i. ■

Lemma D.3. *Multivariate CLT. Let* $E_k = (A_k, B_k)'$, $Var(E_k) = \Sigma_E$ *and* $Z_k = (M_{k,N}, E_k')'$. *Then,*

$$\frac{1}{\sqrt{K}}\sum_{k=1}^{K} Z_k \to N\left(\mathbf{0}, \begin{pmatrix}\theta_N & 0 \\ 0 & \Sigma_E\end{pmatrix}\right)$$

Proof of Lemma D.3. The Z_k vectors are *iid* with mean 0 and finite second moments (by Assumption D.1.ii), therefore standard multivariate CLT applies. The form of variance-covariance matrix is given by Lemma D.2. ■

Lemma D.4. *Approximate Form of* $\hat{b}_K$*.*

$$\sqrt{K}(\hat{b}_K - \theta_{N_p}) = \frac{1}{\sqrt{K}} \sum_{k=1}^{K} B_k + o_p(1)$$

Proof of Lemma D.4.

$$\begin{aligned}
\hat{b}_K - \theta_{N_p} &= \frac{1}{K-1} \sum_{k=1}^{K} \left(M_{k,N_p} - \bar{M}_{K,N_p}\right)^2 - \theta_{N_p} \\
&= \frac{1}{K-1} \left(\sum_{k=1}^{K} M_{k,N_p}^2 - K\bar{M}_{K,N_p}^2 \right) - \theta_{N_p} \\
&= \frac{1}{K-1} \left(K\theta_{N_p} + \sum_{k=1}^{K} (M_{k,N_p}^2 - \theta_{N_p}) - K\bar{M}_{K,N_p}^2 \right) - \theta_{N_p} \\
&= \frac{1}{K-1} \sum_{k=1}^{K} M_{k,N_p}^2 + \frac{\theta_{N_p}}{K-1} - \frac{K}{K-1} \bar{M}_{K,N_p}^2
\end{aligned}$$

Finally,

$$\frac{\sqrt{K}}{K-1}\theta_{N_p} = O(K^{-1/2}) = o(1) \qquad \frac{\sqrt{K}}{K-1} \sum_{k=1}^{K} M_{k,N_p}^2 = \frac{1}{\sqrt{K}} \sum_{k=1}^{K} M_{k,N_p}^2 + o_p(1) \qquad \frac{K\sqrt{K}}{K-1}\bar{M}_{K,N_p}^2 = o_p(1)$$

where the second and third equality follow from the bounded moments assumption in Assumption D.1.ii and noting that $\bar{M}_{K,N_p} = O_p(K^{-1/2})$, therefore $\bar{M}_{K,N_p}^2 = O_p(K^{-1})$, $\frac{K\sqrt{K}}{K-1}\bar{M}_{K,N_p}^2 = O_p(K^{-1/2}) = o_p(1)$. It follows that

$$\sqrt{K}(\hat{b}_K - \theta_{N_p}) = \frac{1}{\sqrt{K}} \sum_{k=1}^{K} B_k + o_p(1)$$

■

Lemma D.5. *Delta Expansion for* $\hat{\lambda}_K$*.*

$$\sqrt{K}(\hat{\lambda}_K - \lambda) = \frac{1}{\sqrt{K}} \sum_{k=1}^{K} L_k + o_p(1)$$

Proof of Lemma D.5. Define the map $g(w, b) = \frac{w}{b - w/N_p}$, then, $\lambda = g(\sigma, \theta_{N_p})$. Now,

$$\nabla g(\sigma, \theta_{N_p}) = \begin{pmatrix} g_w \\ g_b \end{pmatrix} \Bigg|_{w=\sigma, b=\theta_{N_p}} = \begin{pmatrix} \frac{b}{(b-w/N_p)^2} \\ -\frac{w}{(b-w/N_p)^2} \end{pmatrix} \Bigg|_{w=\sigma, b=\theta_{N_p}} = \begin{pmatrix} \theta_{N_p}/\rho^2 \\ -\sigma/\rho^2 \end{pmatrix}$$

By the Delta method,

$$\sqrt{K}(\hat{\lambda}_K - \lambda) = \nabla g(\sigma, \theta_{N_p})' \begin{pmatrix} \sqrt{K}(\hat{w}_K - \sigma) \\ \sqrt{K}(\hat{b}_K - \theta_{N_p}) \end{pmatrix} + o_p(1) = \frac{1}{\sqrt{K}} \sum_{k=1}^{K} L_k + o_p(1)$$

where the equality is exact for $\sqrt{K}(\hat{w}_K - \sigma)$ and $o_p(1)$ as per Lemma D.4 for $\sqrt{K}(\hat{b}_K - \theta_{N_p})$.

■

Corollary D.6. *Asymptotic Measurability of Design. Let $\mathcal{G} = \sigma((L_k)_{k\geq 1})$ be the σ-algebra generated by the $(L_k)_{k\geq 1}$ sequence. TUnder Assumption D.1.iii (K_A, N_A) are $\mathcal{G}$-measurable*

Proof of Corollary D.6. Corollary follows Lemma D.5.

Lemma D.7. *Stable CLT. For some fixed N*

$$\frac{1}{\sqrt{K}} \sum_{k=1}^{K} M_{k,N} \xrightarrow[K\to\infty]{\mathcal{G}\text{-stably}} \sqrt{\theta_N}\, Z, \qquad Z \sim N(0,1),\ Z \perp \mathcal{G},$$

Proof of Lemma D.7. The proof will rely on (random index version of) Martingale Stable CLTs as in Hall and Heyde (2014). To invoke the relevant theorem we need

- Martingale Difference Property: Let $\mathcal{F}_0 = \mathcal{G}$ and $\mathcal{F}_k = \sigma((L_j)_{j\geq 1}, M_{1,N}, \ldots, M_{k,N})$ such that $\mathcal{G} \subseteq \mathcal{F}_k$ for all k. Under cluster independence $\mathbb{E}[M_{k,N} \mid \mathcal{F}_{k-1}] = \mathbb{E}[M - k, N \mid L_k] = 0$ (by Lemma D.2). Thus $(M_{k,N}, \mathcal{F}_k)$ is a martingale difference sequence. By *frontloading* $\mathcal{G}$ in $\mathcal{F}$ we ensure that $\mathcal{F}$-stability implies $\mathcal{G}$-stability.

- Conditional Variance: $\frac{1}{K}\sum_k^K \mathbb{E}[M_{k,N}^2 \mid \mathcal{F}_{k-1}] = \frac{1}{K}\mathbb{E}[M_{k,N}^2 \mid L_k] \to \mathbb{E}[M_{k,N}^2] = \theta_N$ (by the LLN, provided $\theta_N < \infty$). Conditional on N, the variance is constant.

- Lindeberg Condition: Under Assumption D.1.ii, (finite fourth moments) Lyapunov holds, i.e. $\frac{1}{K^2}\sum_k^K \mathbb{E}[M_{k,N}^4] \to 0$. Hence, Lindeberg condition holds.

Under the conditions above (see Hall and Heyde (2014) or Häusler and Luschgy (2015)), by the martingale stable CLT

$$\frac{1}{\sqrt{K}} \sum_{k=1}^{K} M_{k,N} \to N(0, \theta_N), \quad \mathcal{F}\text{-stably}$$

One can further replace K by a random stopping time (in an Anscombe style, see Häusler and Luschgy (2015)) provided $K_A/B \xrightarrow{p} \kappa$ under Assumption D.1). Finally, stable convergence with non-random limiting variance θ_N implies mixing convergence (see Aldous and Eagleson (1978)). And mixing convergence does imply limit independence. ■

Proof of Theorem 6.2. Fix $N \in [N^*(\lambda_m), N^*(\lambda_M)]$. The event $\{N_A = N\}$ is asymptotically $\mathcal{G}$-measurable. Thus, under Lemma D.7

$$\frac{\sqrt{K_A}(\hat{\mu}(K_A - \mu))}{\sqrt{\theta_N}} \to N(0,1), \quad \mathcal{G}\text{-stably}$$

Summing over finitely many $N \in [N^*(\lambda_m), N^*(\lambda_M)]$ and using the fact that $\theta_N \xrightarrow{p} \theta_0$ (provided $N_A \to N^*(\lambda_0)$ under Assumption D.1.iii), recovers the statement in the Theorem. Under Corollary D.6 (K_A, N_A) is $\mathcal{G}$-measurable, so the x-uniform statement follows from Polya's Theorem (with $\Phi(x)$ continuous in x).

The feasible $\hat{b}(K_A, N_A)$ statement follows by noting that $\hat{b}(K_A, N_A) \xrightarrow{p} \theta_0 > 0$ under Assumption D.1.iii. Stable convergence is preserved under multiplication by a sequence converging in probability to a constant therefor,

$$Z_B \sqrt{\theta_0 / \hat{b}(K_A, N_A)} \to N(0,1) \;\; \mathcal{G}\text{-stably}$$

coverage is immediate $\square$